

A multi-scale convolutional LSTM-dense network for robust cardiac arrhythmia classification from ECG signals

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ABSTRACT

Cardiac arrhythmias are irregular heart rhythms that, if undetected, can lead to severe cardiovascular conditions. Detecting these anomalies early through electrocardiogram (ECG) signal analysis is critical for preventive healthcare and effective treatment. However, the automatic classification of arrhythmias poses significant challenges, including class imbalance and noise interference in ECG signals. This paper introduces the Multi-Scale Convolutional LSTM Dense Network (MS-CLDNet) model, an advanced deep-learning model specifically designed to address these issues and improve arrhythmia classification accuracy and other relevant metrics. This paper aims to develop an efficient deep-learning model, MS-CLDNet, for accurately classifying cardiac arrhythmias from electrocardiogram (ECG) signals. Addressing challenges like class imbalance and noise interference, the model integrates bidirectional long short-term memory (LSTM) networks for temporal pattern recognition, Dense Blocks for feature refinement, and Multi-Scale Convolutional Neural Networks (CNNs) for robust feature extraction. To achieve accurate classification of different types of arrhythmias, the Classification Head refines these extracted features even further. Utilizing the MIT-BIH arrhythmia dataset, key pre-processing techniques such as wavelet-based denoising were employed to enhance signal clarity. Results indicate that the MS-CLDNet model achieves a classification accuracy of 98.22 %, outperforming baseline models with low average loss values (0.084). This research highlights how crucial it is to combine sophisticated neural network architectures with efficient pre-processing techniques to improve the precision and accuracy of automated cardiovascular diagnostic systems, which could have important healthcare applications for early and accurate arrhythmia detection.

1. Introduction

Cardiovascular diseases (CVDs) represent a significant worldwide health issue, accounting for a substantial proportion of global mortality, with heart disease and stroke being the leading causes [1]. Early detection of cardiac irregularities is essential for successful treatments and the avoidance of unfavorable outcomes [2]. Diagnostic techniques such as Magnetic Resonance Angiography (MRA) and Electrocardiogram (ECG) are indispensable for identifying cardiovascular problems [3,4]. These technologies help medical professionals detect disorders like arrhythmia, characterized by irregular heart rhythms, which, if untreated, can lead to serious consequences such as stroke and cardiac arrest [5,6]. Thus, precise and effective techniques for arrhythmia

identification are crucial to enhance patient outcomes and reduce the prevalence of cardiovascular diseases worldwide [7,8]. Arrhythmia, an irregular heartbeat, encompasses a range of cardiac rhythm abnormalities, from relatively minor to potentially fatal [9]. Causes include heart illness, electrolyte imbalances, structural anomalies, or adverse drug reactions. Symptoms range from palpitations and light-headedness to dyspnoea and fainting [10]. Treatments include medication, lifestyle modifications, and devices like pacemakers or implanted defibrillators aimed at controlling symptoms and reducing the risk of complications [11,12]. Effective management requires a multidisciplinary approach involving cardiologists, electrophysiologists, and other specialists to tailor interventions to the specific needs and severity of each case [13, 14]. Many advancements in machine learning, particularly deep

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learning, have revolutionized medical diagnostics, offering new possibilities for arrhythmia classification.

The research work by Rajeev Kumar Rai et al. [1] proposes a non-invasive method for monitoring fetal heart activity using abdominal ECG and DenseNet transfer learning, achieving a decent classification accuracy for fetal arrhythmia. Wusat Ullah et al. [2] explored deep learning techniques for arrhythmia classification using datasets like the MIT-BIH Arrhythmia Database, demonstrating high accuracy with models such as CNN + LSTM. Qiao Xiao et al. [3] reviewed 368 studies, highlighting trends like the prevalence of CNNs and the need for advanced denoising methods. Garima Chandel et al. [4] utilized a Linear Discriminant Analysis classifier for real-time arrhythmia detection with high accuracy. Ummay Umama Gronthy et al. [5] conducted a bibliometric analysis identifying key trends in arrhythmia research. Thong Vu et al. [6] proposed a method for real-time arrhythmia detection using CNNs, showing high accuracy and efficiency with wearable sensors. Mengze Wu et al. [7] achieved notable accuracy with a 12-layer CNN for ECG signal classification. Chen et al. [8] combined CNN and LSTM for automated arrhythmia classification. Ojha et al. [9] introduced a CNN-SVM model based on Auto-encoder ConvNet, demonstrating high effectiveness. Midani et al. [10] developed DeepArr, a CNN with BiLSTM for arrhythmia detection. Meng Qi et al. [11] proposed a 9-layer CNN and 5-layer Fully Connected Network, showing notable efficacy. Zhongnan Ran et al. [12] utilized a multi-feature, multi-path parallel deep CNN with LSTM, achieving 91.22 % accuracy under the inter-patient paradigm. As summarized in Table 1 various deep-learning models have been explored for arrhythmia classification. Each study employed different datasets and methodologies for training and validation. For instance, datasets like the MIT-BIH Arrhythmia Database were commonly used, with specific data preprocessing and feature extraction methods tailored to the needs of the respective models. The training procedures varied, with models such as the one by Mengze Wu et al. [7] using 60 epochs and batch sizes of 36 (see Table 1).

In contrast, others like Zhongnan Ran et al. [12] utilized more complex architectures with multiple convolutional layers and LSTMs. The studies demonstrated varying degrees of success. While all the studies in Table 1, did not use the MIT-BIH arrhythmia dataset; their results, as well as the demerits, have been provided for comparison with the proposed model. The outcomes showed that deep learning architectures are helpful for detecting arrhythmias. Still, they also brought to light some drawbacks, including the necessity for big, diverse datasets, overfitting, and high computing costs. For example, Rajeev Kumar Rai et al. [1] achieved high accuracy but faced generalization issues, while Chen et al. [8] attained a notable accuracy but required additional steps like QRS detection.

Building on existing research, this work aims to enhance arrhythmia classification by proposing a Multi-Scale CNN-LSTM-Dense Model integrated with advanced feature extraction techniques. The approach leverages the strengths of deep learning architectures such as DenseNet and combines them with LSTM networks for feature fusion and parallel processing. The objective is to develop a reliable arrhythmia classification system that improves the efficiency and accuracy of arrhythmia detection from ECG signals. The proposed model addresses issues related to dataset diversity, quality, and availability by utilizing multiple ECG databases and applying advanced data preparation techniques. This comprehensive approach ensures a robust architecture capable of accurately classifying a variety of arrhythmic rhythms. By implementing class balancing and ECG signal denoising, the model mitigates challenges associated with imbalanced datasets and noisy signals, which are common in ECG data. This research highlights the importance of effective preprocessing and advanced neural network architectures in improving diagnostic accuracy for cardiovascular health monitoring, ultimately contributing to better patient outcomes and advancements in cardiovascular care.

The proposed MS-CLDNet model introduces several novel contributions that distinguish it from existing models in arrhythmia detection

Table 1

Comprehensive review of existing deep learning models for arrhythmia detection.

S. No	Authors	Technique/Model Architecture	Demerits	Results [Accuracy]
1.	Rajeev Kumar Rai et al. [1]	DenseNet Architecture	Noise-sensitive dataset, data dependency, and generalization concerns	97.56 %
2.	Wusat Ullah et al. [2]	CNN-LSTM, and CNN-LSTM - Attention Model	Overfitting Risk, Computational cost, Large Dataset requirement	• CNN model- 95.12 % • CNN + LSTM- 96.3 % • CNN + LSTM + Attention Model. 96.29 %
3.	Qiao Xiao et al. [3]	CNNs, RNNs, LSTM and BiLSTM	Interpatient-Paradigm, Data Variability, Complex DL models	• O-WCNN- 96.43 % • SE-ResNet- 97.61 % • BiLSTM- 97.53 %
4.	Garima Chandel et al. [4]	Linear Discriminant Analysis (LDA) based classifier	Inconsistencies in classifier performance across different patients	92.57 %
5.	Ummay Umama Gronthy et al. [5]	Bibliometric analysis on Arrhythmia detection	Data Accessibility, methodological constraints	Makes use of relevancy, trend, and citations
6.	Thong Vu et al. [6]	2D CNN network architecture inspired by AlexNet	Data limitations, Hardware constraints	94.7 %
7.	Mengze Wu et al. [7]	A 12-layer CNN Arch was proposed. 60 epochs and batch size of 36. 10-fold cross-validation	Limited dataset size and potential overfitting	97.41 %
8.	Chen C et al. [8]	CNN-LSTM	Low classification accuracy for the atrial flutter (AFL) category, high computing costs, an unbalanced dataset, and QRS detection are requirements of the approach.	97.98 %
9.	Ojha M. K. et al. [9]	CNN-SVM 1D-CNN + Auto-encoder ConvNet	Limited to 1D CNN model and imbalance dataset	97.53 %
10.	Midani W et al. [10]	CNN-BiLSTM and DeepArr [combines feed-forward and recurrent neural networks]	Limited training and testing database, an imbalanced dataset, and R peak segmentation based on dataset annotation were employed.	96.46 %
11.	Meng Qi et al. [11]	9-layer CNN and 5-layer FC Network	Integration of multiple datasets can be potentially biased and computationally complex.	• CNN - 98.01 % • FC - 97.06 %
12.	Zhongnan Ran et al. [12]	Multi-path parallel deep CNN with 20	Imbalanced dataset, complexity of model architecture, and lack of detailed	• intra-patient paradigm - 97.01 %

(continued on next page)

Table 1 (continued)

S. No	Authors	Technique/Model Architecture	Demerits	Results [Accuracy]
		conv layers, LSTM	discussion on training dynamics and convergence	• inter-patient paradigm - 91.22 %

and classification. One of the key innovations is the Multi-Scale Feature Extraction capability. Unlike many models that rely on a single-scale convolutional approach, this model incorporates a Multi-Scale CNN module designed to extract features at various temporal resolutions. The suggested model can capture a wider variety of patterns in ECG data because of this multi-scale approach, making it more robust in detecting subtle arrhythmia features that traditional CNN models may overlook. A Bidirectional LSTM (BiLSTM) layer is integrated, which improves the model's comprehension of temporal dependencies in ECG data. The model's performance was evaluated through a comprehensive experimental setup, including K-fold cross-validation, to ensure robust performance metrics. This rigorous evaluation process contrasts with some studies that may not have employed such thorough validation techniques, ensuring that our model's performance is reliable across different data subsets. The proposed model also incorporates a Dense Block for Feature Refinement following the Multi-Scale CNN module. This Dense Block allows for deeper and more refined feature learning by combining outputs from multiple preceding layers, enabling the model to learn complex representations of ECG data.

This approach contrasts with simpler CNN architectures, such as the one proposed by Mengze Wu et al. [7], which do not benefit from the enhanced feature refinement provided by dense connectivity. In addition to its architectural innovations, the Multi-Scale CNN-LSTM-Dense model is designed with regularization and generalization mechanisms to prevent overfitting.

Methods like regularization strategies and a learning rate scheduler ensure the model's good generalization of new data. This focus on generalization is particularly important when compared to models like that of Ojha et al. [9], which, while achieving high accuracy, may not emphasize generalization as strongly. Finally, the proposed model excels in maintaining high performance across multiple arrhythmia classes. While many existing models achieve high accuracy, the MS-CLDNet model demonstrates consistent performance across all arrhythmia classes, as evidenced by confusion matrices, ROC curves, and precision-recall curves. This consistent performance is critical for real-world applications, where the prevalence of different arrhythmias can vary significantly. Overall, the MS-CLDNet model introduces significant advancements in multi-scale feature extraction, temporal sequence learning, and model generalization. These innovations make it a strong candidate for clinical deployment in arrhythmia detection and classification, setting it apart from existing models.

The key contributions of this paper include.

1. The implementation of class balancing techniques and wavelet-based denoising to address the class imbalance and reduce noise in ECG signals effectively.
2. The creation of a brand-new deep learning architecture for precise arrhythmia classification that combines Dense Blocks, Bi-LSTM, Multi-Scale CNN (MS-CNN), and a Classification Head.
3. Achieving significant improvements in classification accuracy and robustness compared to baseline models.
4. Demonstrating enhanced capability in modeling temporal relationships and capturing multi-scale features, leading to increased diagnostic precision.

The format of this paper is as follows: Section 2 explores the proposed architecture in detail and addresses the databases and

methodology used. Section 3 evaluates the model's performance based on the results. Section 4 concludes with discussions of implications and explores future research directions.

2. Materials and methods

2.1. Description of dataset

A vital aid in cardiovascular research, the MIT-BIH Arrhythmia Database is crucial for the investigation and creation of algorithms aimed at detecting arrhythmias [15,16]. Since its creation, this database has served as a pillar for ECG signal analysis. It consists of 47 subjects' annotated recordings taken between 1975 and 1979. Forty-eight half-hour segments from 24-h ambulatory ECG recordings are included in the dataset. The digitization process covered a 10-mV range and was completed at a speed of 360 samples per second per channel with 11-bit resolution [16–18]. These recordings provide a wide range of cardiac anomalies and cover a broad spectrum of arrhythmic events [19]. Each ECG record in the MIT-BIH database is meticulously annotated, with cardiologist-reviewed markings identifying significant cardiac events such as arrhythmic beats [20]. This high level of detail and accuracy in annotation makes the database an invaluable benchmark for the development and validation of arrhythmia detection and classification algorithms [21]. Its comprehensive nature ensures that researchers can train and test machine learning models on a wide variety of arrhythmic conditions [22]. The different ECG signal classes found in the dataset are shown in Fig. 1 in the form of signals.

In our research, the MIT-BIH Arrhythmia Database plays a critical role. Its detailed and varied recordings provide the necessary foundation for developing our deep-learning model for arrhythmia classification. By utilizing this dataset, we can address significant challenges such as class imbalance and noise interference in ECG signals. The database's diversity allows us to implement and validate preprocessing techniques, including class balancing and denoising, which are crucial for enhancing the quality of ECG signals [23,24]. These improvements enable our proposed MS-CLDNet model to effectively extract, refine, and classify features from ECG signals. The MIT-BIH Arrhythmia Database's vast nature and meticulous annotation guarantee that our model may make notable enhancements in robustness and classification accuracy, underscoring its importance in advancing diagnostic accuracy for cardiovascular health monitoring.

In this paper, a clear separation between training and testing data was made to maintain the integrity of the evaluation process for the MS-CLDNet model. Following a standard procedure, a specific portion of the dataset was allocated for training, while a separate and distinct portion was designated for testing.

2.2. Data preparation techniques

Several preprocessing techniques have been utilized to prepare the ECG signals for effective classification using the proposed MS-CLDNet model [25–27]. These techniques included denoising, z-score normalization, and class balancing. The dataset for this research was split up into three subsets: testing, validation, and training. The training set consists of 15,000 samples, the validation set consists of 5000 samples, and the testing set consists of 5000 samples. A thorough explanation and pertinent equations for the preprocessing techniques are provided in later sections. The signals must be on a consistent scale for the neural network to learn effectively from the data. This technique gives the data a mean of zero and a standard deviation of one. Together, these preprocessing steps significantly improved the signal quality and facilitated better model performance. Incorporating these preprocessing techniques ensured that the input data fed into our deep learning model was of high quality, thereby enhancing the overall effectiveness of our arrhythmia detection system [28,29].

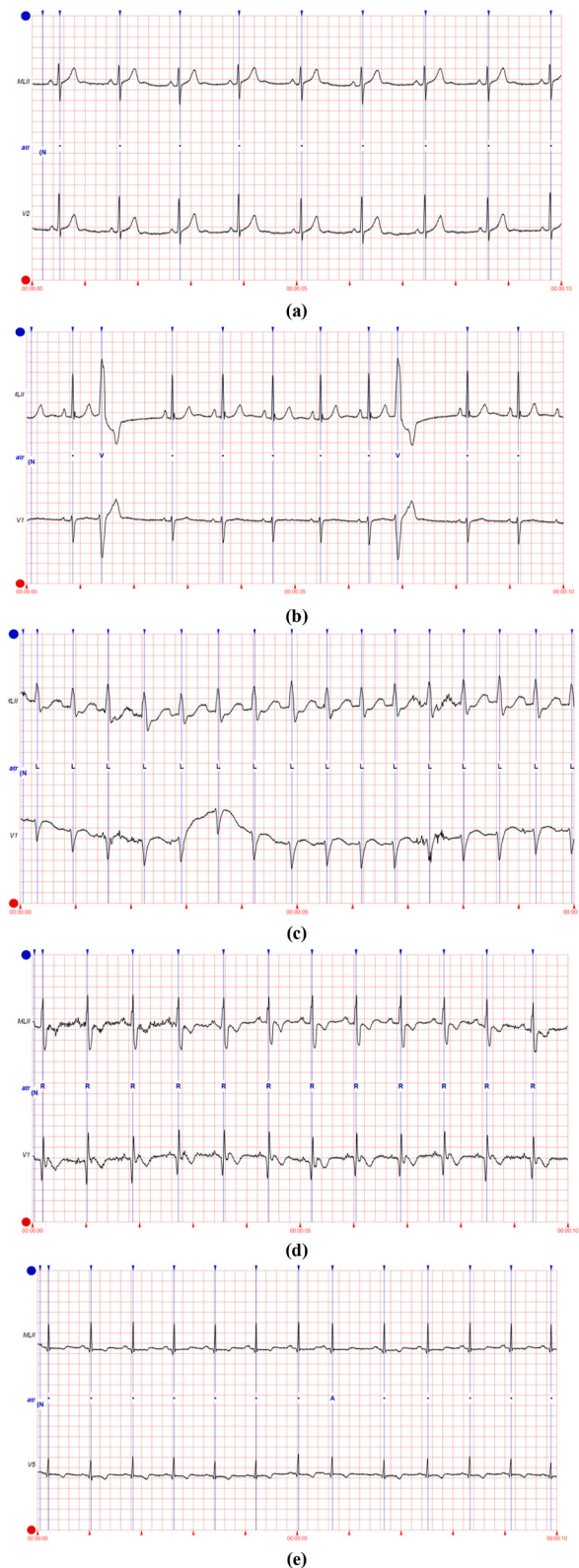


Fig. 1. Different Classes of ECG Signals Present in the Dataset
 (1a) Normal beats (N).
 (1b) V annotations indicating the presence of Premature Ventricular Contractions (PVCs).
 (1c) L annotations, indicating Left Bundle Branch Block (LBBB) beats.
 (1d) R annotations, indicating Right Bundle Branch Block (RBBB) beats.
 (1e) A annotations indicating the presence of Atrial Premature Contraction (APC).

2.3. Class balancing

Class imbalance poses a common challenge in classification tasks, where some classes have notably fewer samples than others [30,31]. This disparity can skew model predictions, as the model might prioritize learning from the majority class, thereby neglecting minority classes.

2.3.1. Quantifying the effects of class imbalance

The dataset exhibits a significant class imbalance, with the majority of samples belonging to class 0 and the minority to class 3. The initial distribution of the dataset is as follows: Class 0: 75,011 samples, Class 1: 8071 samples, Class 2: 7255 samples, Class 3: 2546 samples, Class 4: 7129 samples. This imbalance persists after splitting the data into training and testing sets, as shown in Figs. 3 and 4.

The highest number of samples belong to **Class 0**, while **Class 3** represents the minority class. The dataset was divided into training and testing sets as follows:

This class imbalance affects the model's learning ability, particularly for the minority classes. The imbalance exists exclusively in the target labels (target_train and target_test) and not in the features. The reshaping process only modifies the features, while the labels (train_y and test_y) maintain the same imbalance as before. To analyze the impact of this imbalance, the model was trained with the imbalanced data. The results demonstrate that the model performs well in the majority class but struggles to generalize to the minority classes. The final validation accuracy stabilized at **74.86 %**, indicating that the model's predictive performance is dominated by the majority class, which skews the reported accuracy. A detailed classification report, as shown in Fig. 5, highlights that metrics like precision, recall, and F1-score are significantly lower for the minority classes than for the majority class.

As observed in the classification report, the recall, precision, and F1-score are significantly high for class 0. In contrast, for classes 1, 2, 3, and 4, these metrics are nearly zero, indicating the model's strong bias towards the majority class and its inability to predict the minority classes effectively. The confusion matrix in Fig. 6 further highlights the model's bias towards the majority class (class 0). All predicted values are concentrated in class 0, with no predictions made for classes 1, 2, 3, or 4. It indicates that the model fails to generalize beyond the dominant class, as it does not recognize or correctly classify instances belonging to the minority classes. This result emphasizes the detrimental impact of class imbalance on model performance and the need for techniques to address this issue.

To mitigate this issue, we implemented class balancing techniques, such as oversampling, to ensure an equal contribution from each class during training. We visualized the class distribution using a pie chart to identify any significant imbalances. Addressing these imbalances is crucial to prevent biased predictions and ensure robust model performance across all classes.

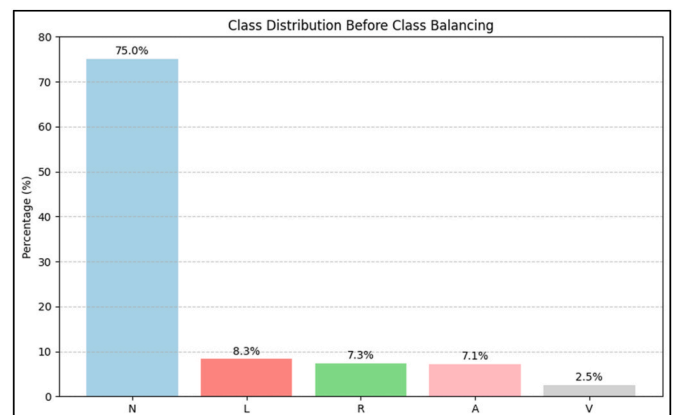


Fig. 2. Actual Class distribution as per in the dataset.

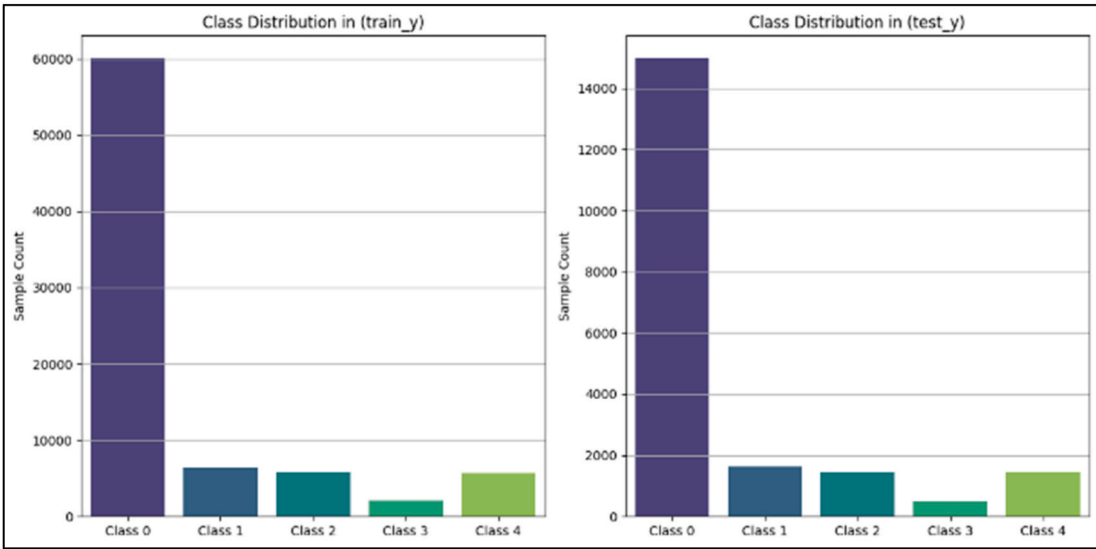


Fig. 3. Visualization of Class imbalance in split Data (test set).

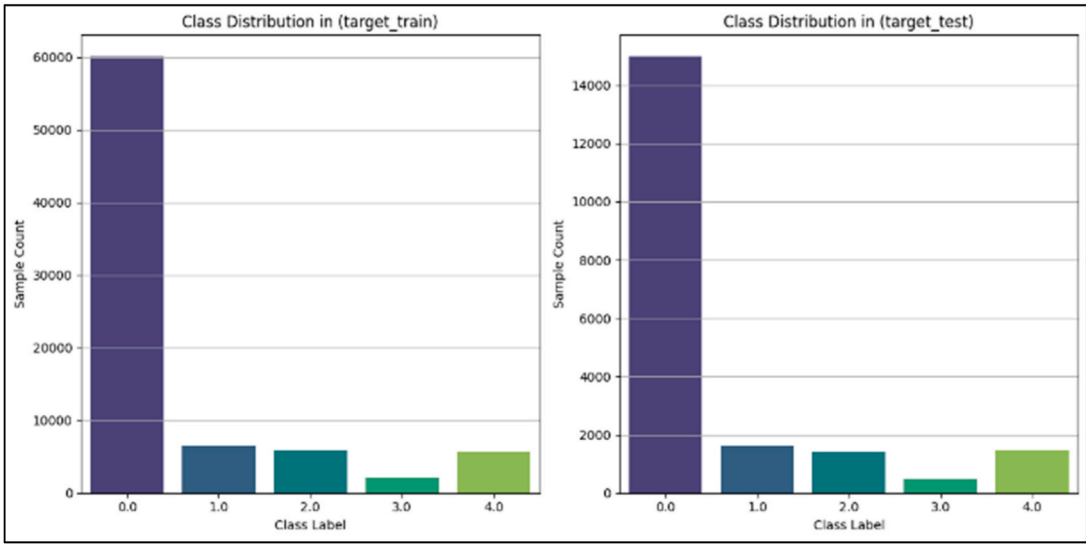


Fig. 4. Visualization of Class imbalance in split data (Train set).

	precision	recall	f1-score	support
0	0.75	1.00	0.86	14975
1	0.00	0.00	0.00	1627
2	0.00	0.00	0.00	1437
3	0.00	0.00	0.00	502
4	0.00	0.00	0.00	1462
accuracy			0.75	20003
macro avg	0.15	0.20	0.17	20003
weighted avg	0.56	0.75	0.64	20003

Fig. 5. Classification Report for the imbalanced dataset.

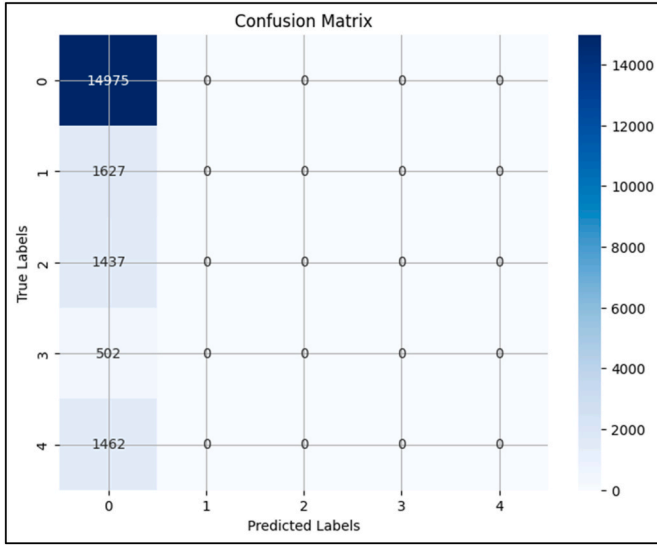


Fig. 6. Confusion matrix for the imbalanced dataset.

A bar chart in Fig. 2 illustrates the significant class imbalance in the MIT-BIH Arrhythmia dataset prior to the application of any class balancing techniques. The "Normal" (N) class overwhelmingly dominates the dataset, comprising 75 % of the total, indicating that most heartbeats are normal. In stark contrast, the "Ventricular" (V) class is the smallest, representing only 2.5 % of the dataset, highlighting the rarity of premature ventricular contractions. The "Atrial" (A) class, accounting for 7.1 % of the dataset, while not as prevalent as regular beats, still represents a notable portion. The "Right Bundle Branch Block" (R) class constitutes 7.3 % of the dataset, similar in frequency to atrial beats.

Meanwhile, the "Left Bundle Branch Block" (L) class, making up 8.1 %, is the second most common after regular beats. This pronounced imbalance underscores the need for class-balancing methods to ensure that machine learning models trained on this dataset can effectively classify all types of heartbeats. The dataset has a significant imbalance, with the normal class vastly outnumbering the other classes. It can pose challenges for ML models, which may become biased towards the majority.

After identifying imbalances, we implemented upsampling of the minority classes using the resampling technique. Random Oversampling was applied to the underrepresented classes (Classes 1, 2, 3, and 4), where instances from these classes were randomly duplicated until their sample size matched that of the majority class (Class 0). This process

ensured that the minority classes were sufficiently represented, allowing the model to learn from all class types more effectively. Additionally, Downsampling was applied to Class 0, the majority class, by randomly selecting a subset of 5000 samples to match the number of instances in the minority classes. The resulting balanced datasets from both over-sampling and downsampling were then concatenated into a single, balanced dataset for training. This approach helped mitigate the class imbalance problem and reduced the risk of model bias towards the majority class. By employing these techniques, the model was exposed to an equal number of samples from each class, thus preventing overfitting and enhancing the model's ability to classify all arrhythmia types correctly. This strategy improves the overall performance of the model, especially for detecting rarer arrhythmia classes, and contributes to more accurate and reliable predictions, as depicted in Figs. 7 and 8. To match the sample count of the majority class, the minority classes' sample counts were artificially increased (oversampled). Fig. 9 represents the balanced classes along with their percentage class distribution. For a class i with n_i samples, the oversampling process is given as follows:

$$X_{-i}^{\hat{}} = \text{resample}(X_i, n_{\text{target}}, \text{replace} = \text{True}) \quad (1)$$

where n_{target} is the desired number of samples per class.

To further address class imbalance in the dataset, additional data augmentation strategies can be employed, such as synthetic sample generation using methods like SMOTE (Synthetic Minority Over-sampling Technique) and its variants. Additionally, applying domain-specific augmentations to the ECG signals, such as time warping, signal flipping, noise injection, and amplitude scaling, can help create diverse training examples while preserving the physiological characteristics of the signals. These techniques can enhance the model's robustness by exposing it to a broader range of variations, thereby mitigating the impact of class imbalance and improving generalization to underrepresented arrhythmia classes.

2.4. ECG signal denoising

In the experiments, a comparison of five different denoising techniques based on several key metrics, including Signal-to-Noise Ratio (SNR), Root Mean Square Error (RMSE), Peak Signal-to-Noise Ratio (PSNR), and Mean Absolute Error (MAE), has been done [32,33]. Table 2 presents a comparison of denoising techniques, including wavelet denoising, moving average, and median filter. Wavelet Denoising was chosen as the best technique for our model. This decision was based on a trade-off between denoising performance and computational efficiency. Wavelet Denoising provided a balanced combination

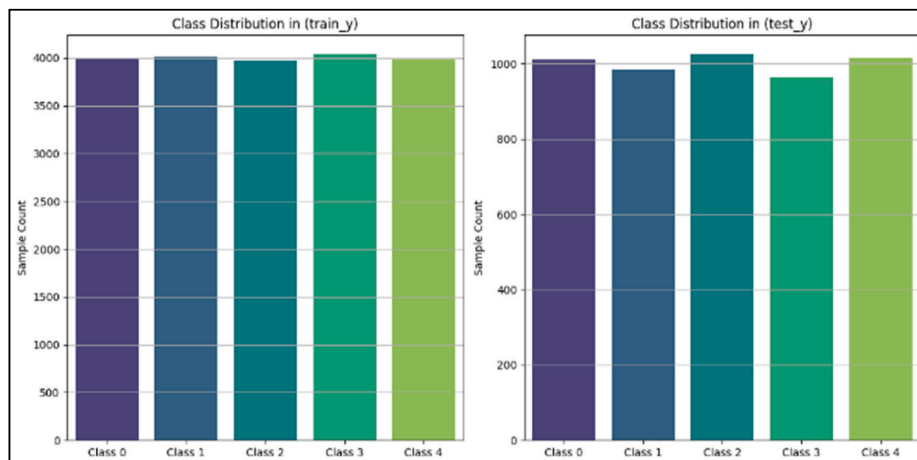


Fig. 7. Visualization of Class Count after balancing the data in split data (Test set).

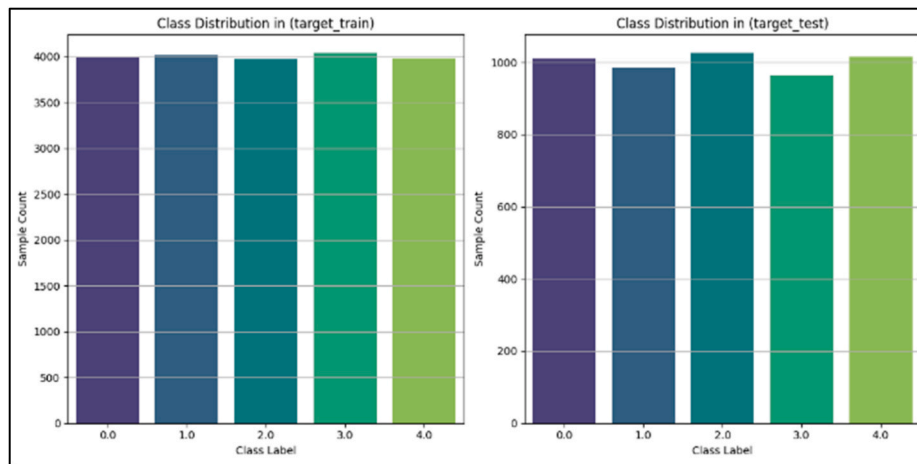


Fig. 8. Visualization of Class Count after balancing the data in split data (Train set).



Fig. 9. Representation of distribution of classes in the dataset after Class balancing.

Table 2

Comparison of denoising techniques.

S.No	Denoising Technique	SNR(db)	RMSE	PSNR (db)	MAE
1.	Wavelet Denoising	11.56	0.26	30.81	0.23
2.	Moving Average	9.28	0.34	28.53	0.12
3.	Median Filter	30.96	0.03	50.22	0.01

of relatively high SNR and PSNR, with acceptable RMSE and MAE values [34,35]. More importantly, it performed well in terms of preserving signal characteristics, which is essential in the context of ECG signal analysis, where preserving original features while removing noise is critical for accurate classification. Additionally, Wavelet Denoising was chosen due to its widespread application in the literature, particularly for ECG signal processing. Its ability to handle non-stationary and noisy ECG signals while maintaining signal integrity has been well-documented, reinforcing its selection for this study [36,37].

A wavelet-based denoising technique was employed as part of the preprocessing of ECG signals to remove noise from the signals while preserving the underlying signal structure [38]. The ECG signals were broken down into wavelet coefficients for the denoising process. The noise components were subtracted using a threshold, and the denoised signals were then reconstructed using the updated coefficients [39,40]. We improved the quality of the input data by denoising the ECG signals, which improved the model's performance and accuracy in classifying arrhythmias [16]. The original signal $x(t)$ is decomposed into wavelet

coefficients $c_{j,k}$ using a wavelet transform. Usually, this decomposition consists of an approximation coefficient $a_{j,k}$ at each level j and several levels of detail coefficients $d_{j,k}$. The detail coefficients $d_{j,k}$ undergo a thresholding operation to eliminate noise following decomposition. The detail coefficients are subjected to an element-wise application of the thresholding function T , as outlined in Equation (2) [41,42].

$$\text{coefficients}_{\text{thresh}} = T(\text{coefficients}, \text{threshold}) \quad (2)$$

The thresholding function can take various forms, such as hard thresholding, soft thresholding, or other adaptive thresholding methods. Equation (3) describes how the denoised signal $x_{\text{denoised}}(t)$ is reconstructed from the modified coefficients using an inverse wavelet transform [13,14].

$$x_{\text{denoised}}(t) = \sum_{j,k} \text{coefficients}_{\text{thresh}}(j,k) \psi_{j,k}(t) \quad (3)$$

where $c_{j,k}$ are the denoised wavelet coefficients.

The ECG signal is compared in the two figures, Fig. 10(a) and (b), before and after a denoising technique is applied. The ECG signal makes a lot of noise before denoising, which is especially noticeable in the baseline fluctuations and the smaller peaks and valleys spread throughout the signal. It is more difficult to distinguish the P waves, QRS complexes, and T waves—three important components of the ECG signal—when high-frequency noise components are present. The ECG signal looks considerably cleaner after denoising, with the high-frequency noise components significantly diminished or eliminated. The main characteristics of the ECG signal are more pronounced and straightforward to recognize, particularly the QRS complexes.

Additionally, the baseline is smoother, allowing for better interpretation of the underlying cardiac activity. Due to the more precise representation of the signal's essential characteristics, this improvement in signal quality improves the detection of significant features, which is essential for the reliable classification of arrhythmias. It also enhances the performance of machine learning models used for the detection and classification of arrhythmias. Overall, the denoising technique has successfully improved the clarity and usability of the ECG signal, making it more suitable for further analysis and classification tasks.

2.5. Z-SCORE normalization

After denoising, we applied z-score normalization to standardize the ECG signals. By adjusting the data to have a 0 mean and a 1 standard deviation, this normalization technique helps the neural network perform better. The z-score normalized signal z for a signal x with mean μ and standard deviation σ is as follows:

$$z = (x - \mu) / \sigma \quad (4)$$

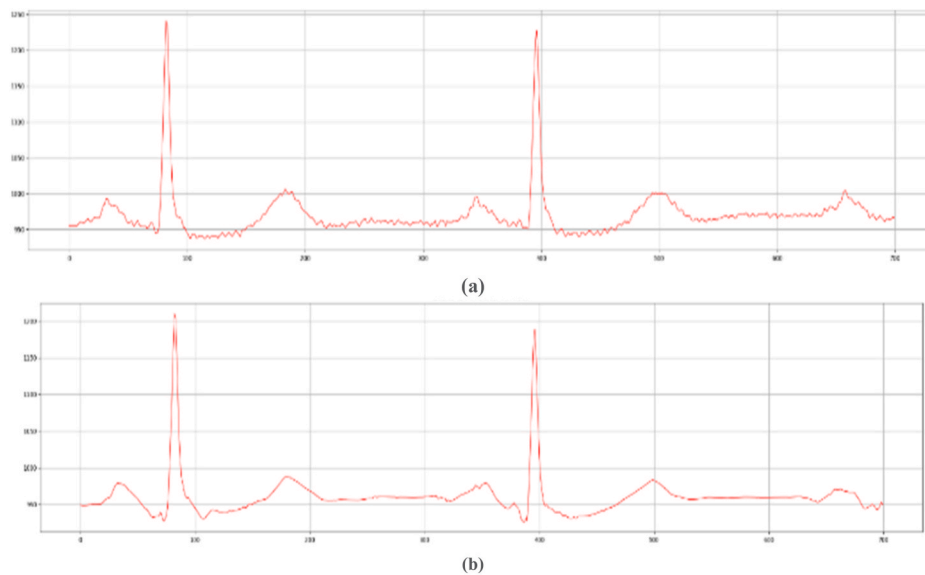


Fig. 10. Ecg Signal waves before and after Denoising

(10a) original ECG Signal with Visible noise, including high-frequency components and baseline Fluctuations.

(10b) ECG Signal after the application of wavelet-based denoising technique.

This process ensures that the signals are on a consistent scale, which is crucial for the neural network to learn from the data effectively. The output of z-score normalization is a set of ECG signals, as shown in Fig. 11(a), where all values are relative to the dataset's mean and standard deviation, eliminating the influence of varying signal magnitudes and providing a uniform input for the model. Together, these preprocessing steps significantly improved the signal quality and facilitated better model performance. Fig. 11(b) illustrates a beat from the example signal after wavelet denoising and z-score normalization.

The flow of signal pre-processing techniques is illustrated in Fig. 12. Denoising involves using a wavelet-based method to remove noise from the signals, enhancing the clarity and quality of the ECG data. By breaking down the signal into its constituent frequency components, this method permits the separation and elimination of noise while preserving crucial cardiac characteristics. Z-score normalization was used after denoising to standardize the signals. The R-peaks were detected by analyzing the denoised and normalized signals, allowing for effective

extraction and classification of features. The ECG signals produced by the preprocessing stages were clear, balanced, and standardized, which was essential for enhancing the functionality of our MS-CLDNet model. The neural network could efficiently learn from the data without being impacted by noise or scale variations thanks to the denoised and normalized signals. By using the class balance strategy, the model was kept from favoring the majority class, leading to more accurate and robust arrhythmia classification across all classes.

2.6. MS-CLDNet model for arrhythmia classification

The research suggests a novel hybrid deep learning model architecture for the classification of arrhythmias from ECG signals. The model combines several neural network components to take advantage of each one's unique capabilities when processing sequential data—more precisely, the 1D time-series data that is commonly found in ECG recordings. Several essential modules make up the overall suggested

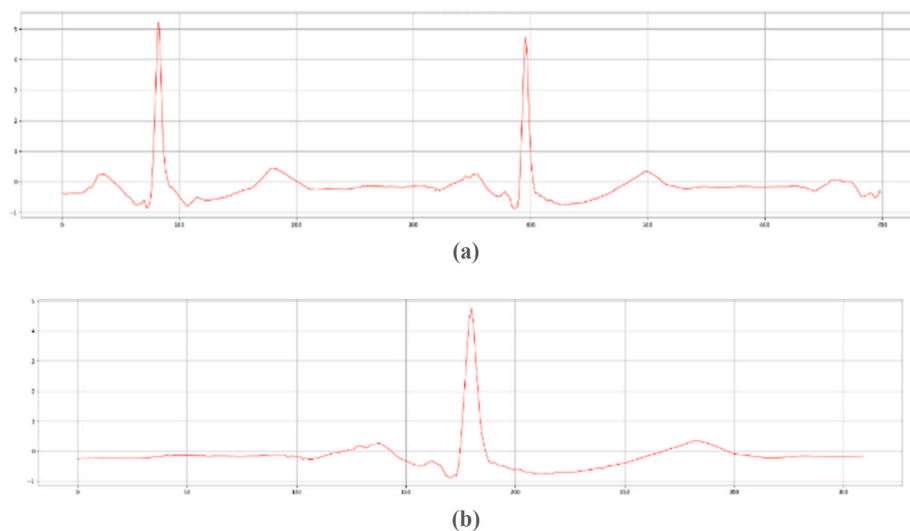


Fig. 11. Wave and Beat After Z-score Normalization

(11a) Wave After z-score Normalization.

(11b) Beat After z-score Normalization.

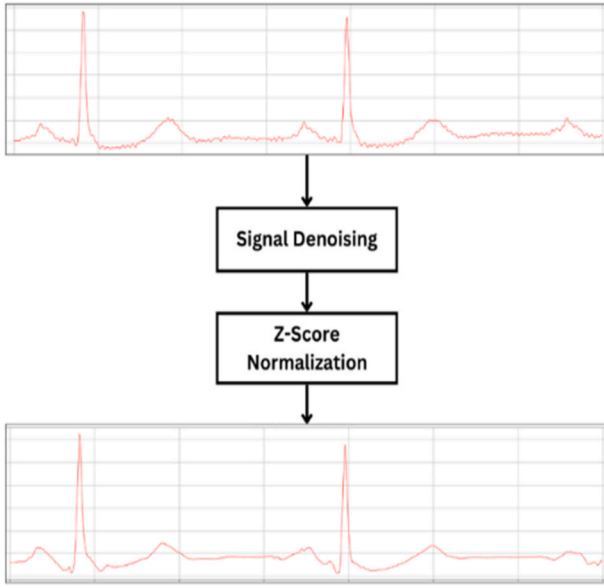


Fig. 12. Flowchart of Signal pre-processing.

architecture, known as MS-CLDNet: an MS-CNN, a Dense Block, a Bi-LSTM network, and a Classification Head made up of numerous dense layers and dropout layers. Each module is specifically tailored to extract, refine, and classify features from ECG data. The MS-CNN module captures features at multiple temporal resolutions from the ECG signal input, which has a shape of (360, 1). To extract features at different scales, this module comprises three concurrent 1D convolutional layers with varied kernel sizes. The first convolutional layer contains 64 filters with a kernel size of 5, the second layer comprises 64 filters with a kernel size of 7, and the third layer contains 64 filters with a kernel size of 9. Every one of these layers employs the same ReLU activation function and padding to maintain the input dimensionality. The output, y_i , of each convolutional layer is computed as follows.:

$$y_i = \text{ReLU}(\text{Conv1D}_{(k,i)}(x)) \quad (5)$$

where $\text{Conv1D}_{(k,i)}$ represents the 1D convolution with kernel size k . The concatenated output is batch-normalized to form a

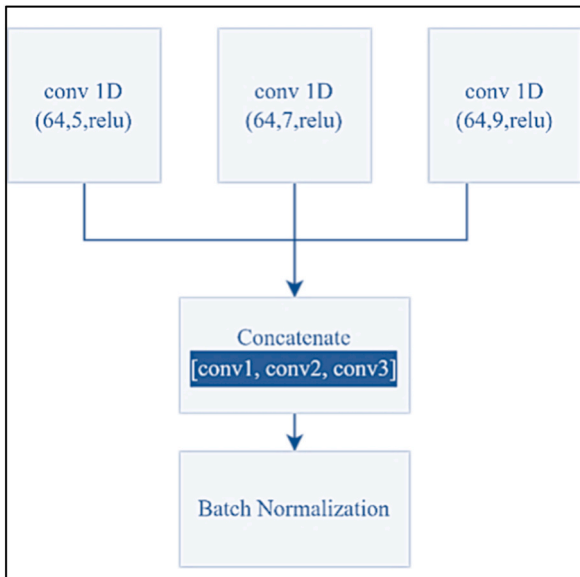


Fig. 13. Multi-scale Convolution neural network architecture.

comprehensive feature representation. This architecture, as illustrated in Fig. 13, effectively captures diverse patterns and features present in the ECG signal.

The Dense Block refines and enhances the extracted features. Several smaller convolutional blocks make up this structure. Each block within this module includes the following operations.

- A one-dimensional convolutional layer with the same padding, ReLU activation, kernel size of 1, and growth rate multiplied by four.
- Batch normalization to stabilize and accelerate training.
- Following batch normalization, there is one more 1D convolutional layer with a growth rate, a kernel size of 3, ReLU activation, and the same padding.
- Each block's output x_{out} is concatenated with the block's input x to ensure feature reuse:

$$x_{out} = \text{Concat}(x, \text{BatchNorm}(\text{ReLU}(\text{Conv1D}_{(3)}(\text{BatchNorm}(\text{ReLU}(\text{Conv1D}_{(1)}(x)))))) \quad (6)$$

The combination of these layers and operations within the Dense Block ensures that the features are effectively refined, allowing the model to capture complex patterns within the data. Each convolutional block's output is concatenated with its input, allowing for feature reuse and mitigating the vanishing gradient problem. Our model uses four such blocks, leading to a highly dense feature representation. The model uses four such dense blocks, leading to a highly dense feature representation, as illustrated in The Bi-LSTM module is intended to record the ECG signal's temporal relationships, as seen in Fig. 14. The model can analyze the sequence both forward and backward by utilizing a bi-directional LSTM with 64 units, which allows it to extract extensive contextual information from the time-series data. The construction of this model is depicted in Fig. 15.

High-level features are mapped to the output classes by the classification head. It is made up of several Dense layers.

- The first Dense layer has 128 units with ReLU activation.
- The second Dense layer has 64 units with ReLU activation.

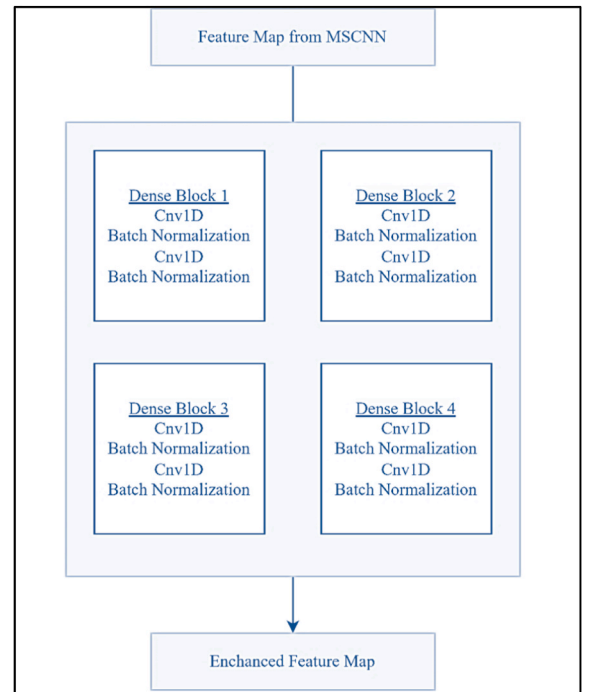


Fig. 14. Internal dense block architecture for feature refinement and enhancement.

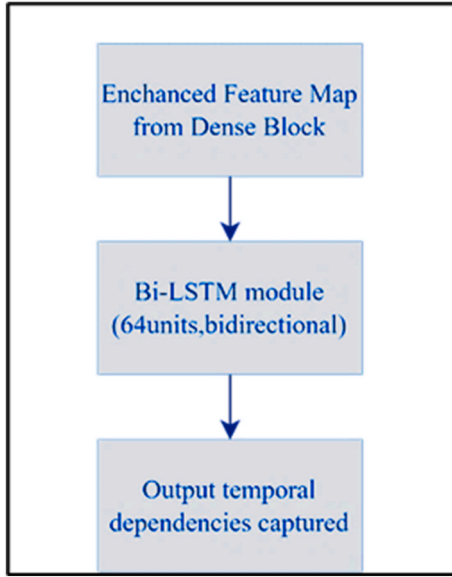


Fig. 15. Bi-LSTM module for temporal dependency capture.

- The final Dense layer outputs the probability distribution across the five arrhythmia classifications using a softmax activation function.
- The classification head's output, y , is calculated using Equation (7).

$$y = \text{Softmax}(\text{Dense}_{64}(\text{ReLU}(\text{Dense}_{128}(\text{Flatten}(h)))))) \quad (7)$$

The model architecture of our proposed model is illustrated in Fig. 16. The overall MS-CLDNet model integrates these modules as follows.

- The input ECG signal is passed through the MS-CNN module to extract multi-scale features.
- The resulting feature map is processed through the Dense Block to enhance feature representation.
- To capture temporal dependencies, the Bi-LSTM module receives the output from the Dense Block.
- Lastly, the Classification Head processes the output from the Bi-LSTM module to produce the final class predictions.

To create a strong model for arrhythmia classification, this multi-faceted strategy makes use of the advantages of convolutional neural networks for feature extraction, dense blocks for feature improvement, and LSTM networks for temporal dependency capture.

The Multi-Scale CNN component extracts features from the input ECG signal at different temporal resolutions, which is crucial for addressing noise interference and improving model robustness in the context of arrhythmia detection. The model applies three parallel convolutional layers with different kernel sizes (5, 7, and 9), which helps to capture multi-scale temporal patterns in the ECG signal. This approach ensures that the model can detect arrhythmic events at varying

granularities, from short-term fluctuations to longer-term trends. By capturing features at multiple scales, the model is less susceptible to noise artifacts that may affect only specific frequency ranges, ensuring more accurate feature extraction even in the presence of signal distortions. The multi-scale approach helps the model become more sensitive to various arrhythmic patterns, including rare or less represented classes. The Dense Blocks play a critical role in improving feature extraction by encouraging the model to learn and retain complex patterns from the ECG signals. Each dense block consists of multiple convolutional operations that increase the feature richness, ensuring that the model can capture intricate patterns associated with arrhythmia events. The dense connections in the dense block help propagate important information throughout the network, ensuring that key features are not lost, even in the presence of noise. After feature extraction through MS-CNN and temporal sequence learning with Bi-LSTM, the model uses a fully connected classification head to make the final predictions.

In this paper, a Bi-LSTM (Bidirectional Long Short-Term Memory) network is used to capture temporal dependencies in the arrhythmia classification task, where the sequences of electrocardiogram (ECG) signals contain crucial patterns that span both past and future time steps. The Bi-LSTM has the advantage of processing sequences in both forward and backward directions, making it particularly valuable for ECG signals, where the context of both preceding and subsequent beats significantly contributes to the correct classification of arrhythmias. Arrhythmias often manifest in specific patterns over time, where both previous and subsequent heartbeats influence the prediction of the next heartbeat or arrhythmia event. These temporal patterns can provide additional context, making it easier to differentiate between normal and abnormal heart rhythms. Arrhythmias can have irregular patterns that may involve complex relationships between successive heartbeats. For example, an irregular rhythm might require considering not just the previous heartbeat but also upcoming beats to properly characterize its behavior (e.g., premature ventricular contractions or atrial fibrillation). A Bi-LSTM captures both past and future contexts, making it well-suited for these types of sequential dependencies. Some arrhythmic events can have both early and late phases that are critical for classification. For instance, early signs of an arrhythmia might not be as distinct, but when combined with future beats, they provide a clearer indication of abnormal rhythm. It is where the future context becomes valuable—providing a fuller picture of the ongoing heart activity, which helps in accurately classifying the arrhythmia.

The implementation of the proposed model begins by preparing the input data, 'X' (features) and 'y' (target), extracted from a DataFrame `X_train_df`. The target 'y' is transformed into a multi-class classification-ready categorical format. The model architecture is defined by multiple key functions: `multi_scale_cnn`, which uses 1D convolutional layers with different kernel sizes to extract multi-scale features from ECG signals; `dense_block`, which improves feature representation through dense convolutional blocks; `bi_lstm`, which uses a bidirectional LSTM network to capture temporal dependencies; and `classification_head`, which uses dense layers with softmax activation for final classification. These functions are integrated into the `ms_cnn_lstm_dense` function, which assembles the complete model. Using the Adam optimizer, categorical cross-entropy loss function, and accuracy metric, the `create_model`

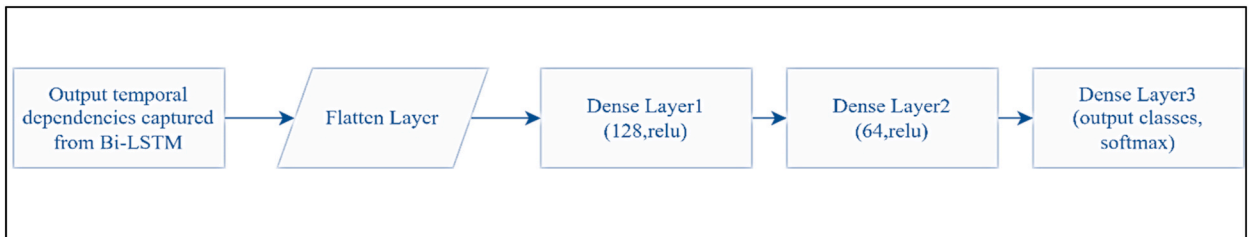


Fig. 16. Classification head for mapping features to output classes and integrated MS-CLDNet model architecture.

function assembles the model. To track the learning rate's evolution during training, a custom callback 'LrHistory' records the learning rate at the end of each epoch. A learning rate scheduler 'scheduler' is implemented to decrease the learning rate exponentially after an initial phase, optimizing model training efficiency. For robust evaluation, the algorithm implements K-fold cross-validation with 'n_splits = 5', ensuring the dataset is split into five folds for train and val. Every fold iteratively trains the model on a subset ('train_x', 'train_y') and evaluates it on the remaining subset ('test_x', 'test_y'). With a batch size of 64, training proceeds over nine epochs while making use of the specified callbacks for history tracking and LR adjustment. Metrics of the model's performance, like loss and accuracy, are assessed and recorded after every fold is trained. The average accuracy and loss of overall folds are calculated by averaging the results from each fold, which reflects on the model's performance consistency and capacity for generalization. The final step of the algorithm prints these combined metrics and the standard deviation to evaluate variability between folds and validate the model's ability to identify arrhythmias from ECG signals. This structured approach ensures thorough evaluation and optimization of the MS-CLDNet model, leveraging K-fold cross-validation and learning rate scheduling to enhance robustness and accuracy in clinical decision-making scenarios. Adjustments can be made to hyperparameters and model configurations based on specific data characteristics and performance requirements. The final model is depicted in Fig. 17.

2.7. Impact of noise interference and noise robustness of the proposed MS-CLDNet model

2.7.1. Noise-augmented dataset preparation and model training

To simulate real-world scenarios and evaluate model robustness, a noise-augmented dataset was prepared, including varying levels of noise. The first step involved is signal loading, where signals are extracted from the provided records and annotated files. Following this, denoising was applied using a custom denoising function to remove baseline wander and other artifacts from the signals. To ensure uniformity across samples, the signals were then normalized using z-score normalization.

In the noise augmentation phase, Gaussian noise was introduced to the signals at varying Signal-to-Noise Ratio (SNR) levels to mimic real-world noise scenarios. A function was designed to add Gaussian noise based on the desired SNR levels (5 dB, 10 dB, 20 dB, and 30 dB). The noise was added at four distinct SNR levels, representing increasingly cleaner conditions. For each noisy signal, R-peak annotations were used to extract fixed-length segments, or beats, around the R-peaks. These beats were then used as input data for model training. Example visualizations of a noisy signal and its corresponding noisy beat segment are shown in Fig. 18.

Class balancing was also an important consideration in the preparation of the dataset. The distributions of different arrhythmia classes were monitored throughout the process to prevent overfitting dominant arrhythmia types. To achieve balance across the dataset, the counts of each arrhythmia class were capped. The final dataset included both clean and noisy beats, with noise added at multiple SNR levels. This augmentation resulted in a significantly expanded dataset, with an input feature matrix shape of (400048, 360) and target labels with a shape of (400048) representing the inclusion of four different noise levels. For model training, the complete augmented dataset was used. The model architecture used for training is the proposed MS-CLDNet, loss function, optimizer, and learning rate, which were carefully chosen to optimize performance. The dataset was split into training, validation, and testing sets, ensuring equal representation of noise levels across these sets.

3. Results and analysis for the noise-augmented dataset

The performance of the model on the noise-augmented dataset is promising, as shown by the classification report in Fig. 19 and the

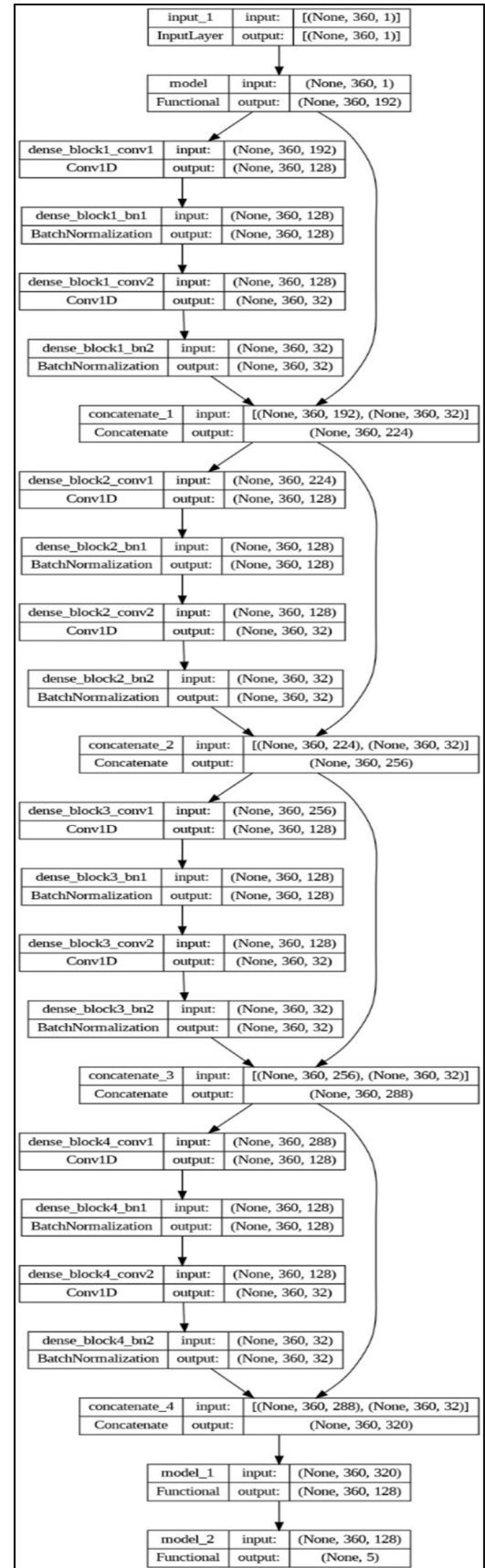


Fig. 17. Complete model architecture of the proposed model, MS-CLDNet model integrated with feature extraction and temporal context.

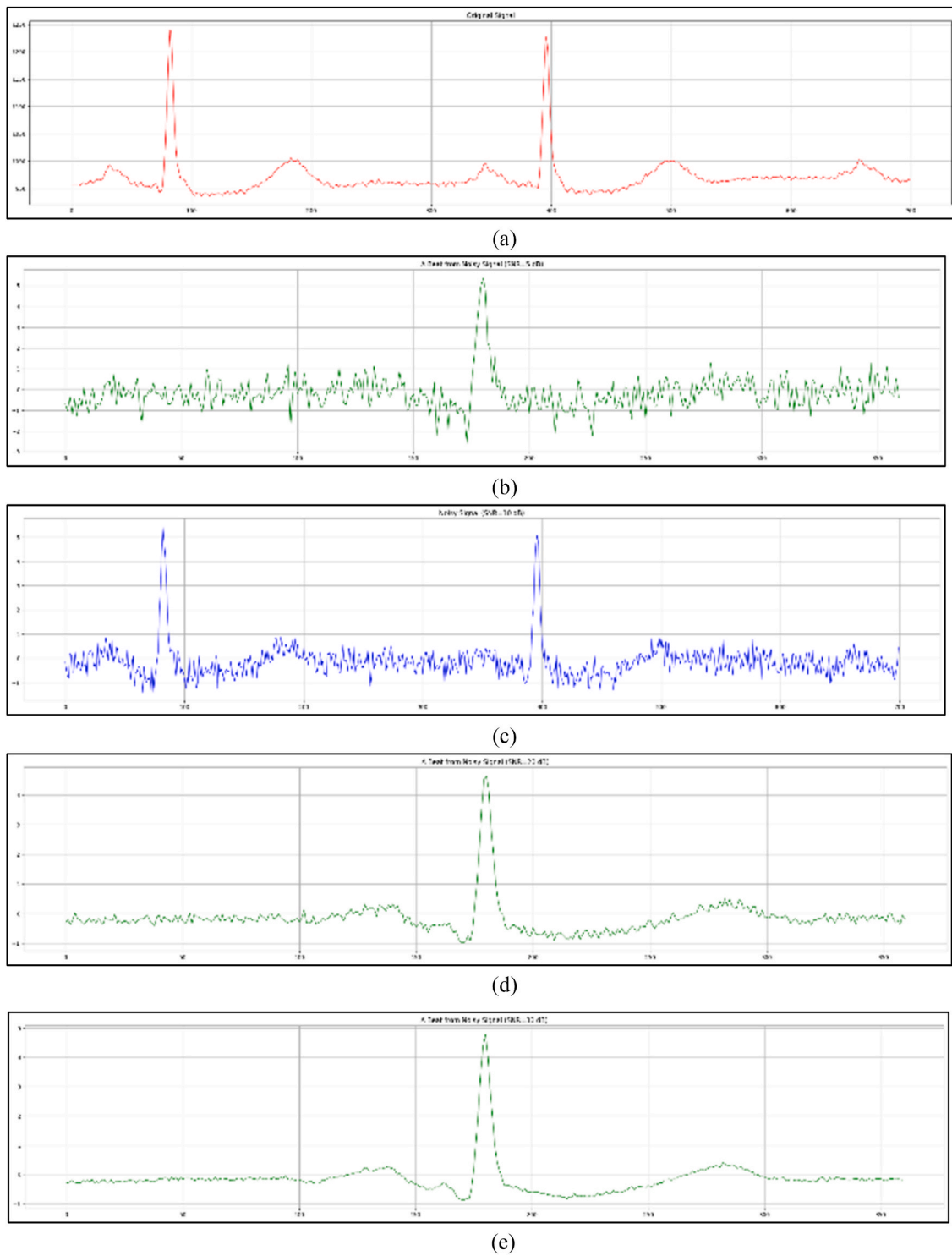


Fig. 18. A sample ECG Signal Wave after embedding varying levels of noise in SNR range [5 dB,10 dB,20 dB,30 dB]

(18a) A sample Original ECG Signal

(18b) ECG Signal with SNR = 5 dB noise level

(18c) ECG Signal with SNR = 10 dB noise level

(18d) ECG Signal with SNR = 20 dB noise level

(18e) ECG Signal with SNR = 30 dB noise level.

confusion matrix represented by Fig. 20, with an overall accuracy of 96.26 %. This high accuracy demonstrates the model's ability to effectively classify the signals, even with the added Gaussian noise at various SNR levels. The precision, recall, and F1 score metrics across all classes further substantiate the robustness of the model, indicating that it is

well-balanced and performs effectively across different arrhythmia classes. The model shows high precision across all classes, with values ranging from 0.91 to 0.99. The recall values, which represent the proportion of true positive predictions among all actual positive samples, range from 0.95 to 0.96 for most classes, indicating that the model is

	precision	recall	f1-score	support
0	0.91	0.95	0.93	1006
1	0.99	0.97	0.98	994
2	0.98	0.97	0.98	1046
3	0.94	0.95	0.95	967
4	0.99	0.96	0.98	987
accuracy			0.96	5000
macro avg	0.96	0.96	0.96	5000
weighted avg	0.96	0.96	0.96	5000

Fig. 19. Classification Report for Noise-Augmented Dataset case.

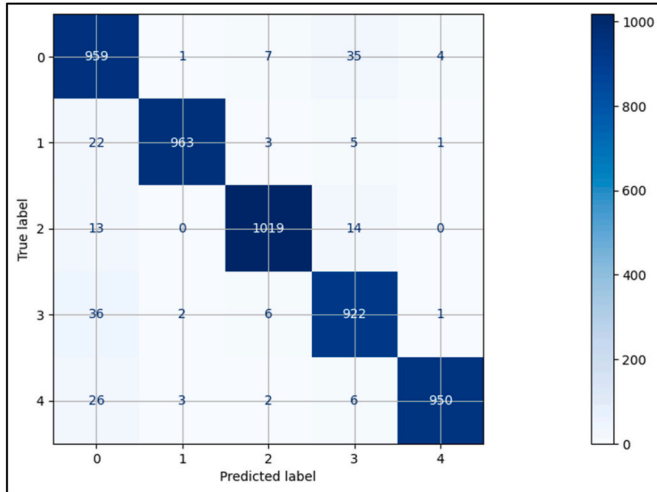


Fig. 20. Confusion Matrix for Noise-Augmented Dataset case.

effective at identifying the majority of samples from each class. Class 1 exhibits the highest recall, while Class 3 shows a slightly lower recall value, implying that while the model is very successful at detecting most signals from Class 1, it might slightly miss some of the signals from Class 3. Class 1 achieves the highest F1 score, demonstrating a strong balance between precision and recall. Nonetheless, the overall performance remains strong across all classes, with the F1 scores indicating that the model is effective in making well-rounded predictions without significant bias toward any particular class. The macro average and weighted average F1 scores confirm that the model performs consistently across different classes without favoring any specific class in the dataset. Table 3 summarizes the precision, recall, and F1 score for each class.

In conclusion, the results indicate a strong ability to classify arrhythmia types under varying noise conditions. The high precision, recall, and F1 scores across all classes suggest that the noise-augmented dataset has helped the model generalize well to real-world scenarios. However, there is slight room for improvement in Class 3, where the recall and F1 scores are marginally lower. Optimization of the SNR levels might enhance performance, especially for this class.

Table 3
Evaluation of Metrics Class wise for the Noise Augmented Dataset case.

Class	Precision	Recall	F1 Score
0	0.908	0.953	0.930
1	0.994	0.969	0.981
2	0.983	0.974	0.978
3	0.939	0.953	0.946
4	0.994	0.963	0.978

3.1. Hyperparameter optimization for the proposed MS-CLDNet model

A systematic hyperparameter tuning process was conducted to optimize the performance of the Multi-Scale Convolutional LSTM Dense Network (MS-CLDNet). The goal was to identify the best combination of hyperparameters to achieve the highest accuracy and minimize loss. A cyclic learning rate schedule was implemented, varying the learning rate between 0.001 and 0.01 over a span of 100 epochs. The optimal learning rate was determined to be 0.005, based on the lowest validation loss observed during the tuning process. Several batch sizes (16, 32, 64, and 128) were tested, with batch size 32 being selected as the optimal value. This choice struck the best balance between GPU memory usage and model performance. A dropout rate of 0.3 was found to provide the best performance, offering a balance between regularization and model accuracy. The number of LSTM units varied between 32 and 128. The optimal configuration was found to be 64 units, based on the evaluation of performance metrics across different configurations. The growth rate of the DenseNet architecture of 24 was chosen, as it showed the best results in terms of model accuracy and convergence speed. Table 4 presents the tuning process and how each hyperparameter choice impacted the model's accuracy.

4. Results and analysis

The proposed model, denoted as MS-CLDNet, combines the advantages of Multi-Scale Networks (MSA-CNN), Bi-LSTM layers, and Dense layers to detect and classify arrhythmias from ECG data effectively. The model is made to take advantage of the temporal sequence learning of LSTMs, the multi-scale feature extraction capabilities of CNNs, and the dense layers' ability to learn complex representations. We trained, tested, and assessed our model using the MIT-BIH Arrhythmia dataset, a standard benchmark for arrhythmia detection and classification tasks.

The research employed Google Colab with GPU acceleration to develop and evaluate a Multi-Scale CNN-LSTM-Dense model for arrhythmia classification using the MIT-BIH dataset. The dataset was divided into a ratio of 70 %, 15 %, and 15 % for training, validation, and test sets, respectively, ensuring sufficient data for training and evaluation while maintaining generalizability. The model architecture comprised a Multi-Scale CNN module for feature extraction at multiple temporal resolutions, followed by a Dense Block for feature refinement and a Bi-LSTM network to capture temporal dependencies in ECG signals. The Classification Head consisted of densely connected layers for the final classification into five arrhythmia classes. The training process employed an Adam optimizer with a learning rate scheduler, reducing the learning rate exponentially after the initial epochs to fine-tune model convergence. K-fold cross-validation (K = 5) was utilized to validate model performance across different folds, ensuring robustness and reliability. Performance metrics, including accuracy and loss, were evaluated on the test set to assess the model's effectiveness in clinical arrhythmia detection tasks.

Equations (8)–(11) were used to calculate Accuracy, precision, recall, and AUC-ROC, respectively.

Table 4
Hyperparameter tuning results and impact on model accuracy.

Hyperparameter	Search Space	Optimal Value	Impact on Accuracy (%)
Learning Rate	0.001 to 0.01	0.005	+1.2
Batch Size	16, 32, 64, 128	32	+1.1
Dropout Rate	0.2 to 0.5	0.3	+0.8
LSTM Units	32, 64, 128	64	+0.9
DenseNet Growth Rate	12, 24, 32	24	+1.3

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

$$\text{AUC - ROC} = \int_0^1 (TP(FPR)dFPR) \quad (11)$$

As can be seen from the training vs. test loss plot in Fig. 22, the model successfully learns from the training data, as seen by the training loss of the model reducing across epochs. However, the test loss varies, indicating that the model may be overfitting to the training set and underfitting to the test set, which contains cases that have not yet been observed. Similarly, the training vs. test accuracy plot depicted in Fig. 21 showcases the model's strong learning ability. Training accuracy steadily climbs to near perfection, indicating the model excels at capturing patterns within the training data. Encouragingly, test accuracy follows a similar upward trend, reaching a high value, which suggests the model generalizes well to the test data. However, the close alignment between training and test accuracy throughout training might hint at some potential overfitting. It could be addressed with techniques like regularization for further improvement. Overall, both the plots indicate the model's ability to learn and perform well on both familiar and new data.

The training vs. test loss plot, as shown in Fig. 22, demonstrates that the model has effectively learned from the training data, with the training loss declining across epochs. However, if the test loss fluctuates, the model can be overfitting to the training data and not generalizing adequately to new cases (test data). The model's strong learning ability is also demonstrated by the training vs. test accuracy plot shown in Fig. 21. The model is good at detecting patterns in data, as evidenced by the training accuracy that increases gradually until it approaches perfection. Test accuracy, on the other hand, shows a positive trend and reaches a high value, indicating that the model performs well in terms of generalizing to the test data. Nonetheless, possible overfitting may be indicated by the close alignment of test accuracy during training with training. Regularization techniques could be used to address this and improve further. The two plots show how well the model can learn from both new and known data.

The convergence of training and validation losses over the epochs is plotted in Fig. 23. Consistent decreases in the training loss show that the model is picking new skills and becoming more proficient with the training set. The validation loss initially exhibits fluctuations but follows a similar trend, suggesting small overfitting as the model tries to generalize to unknown data. The model starts well in the first epoch, as evidenced by the low initial training loss (0.0113), but the higher validation loss (0.0647) is expected because the model has not fully learned the patterns. Due to model modifications, training loss slightly increases over epochs 2 to 4 but then continues to decline, suggesting a general

learning trend. The model's attempts to generalize are reflected in the fluctuations in validation loss. A significant increase in validation loss (0.1026) and training loss (0.0267) during epochs 5 and 6 points to an overfitting issue, where the model might be learning patterns from training data by memory rather than generalizable features. However, from epochs 7 to 9, the training loss sharply decreases, reaching 0.0016, indicating significant learning. Although the validation loss also reduces, it fluctuates, showing the challenge of achieving perfect generalization. The model's ability to extract features from ECG data, a crucial step in the process of identifying arrhythmias, is indicated by the consistently low training and validation loss values. However, the observed fluctuations in validation loss indicate areas for further optimization, such as hyperparameter tuning or regularization techniques to enhance generalization and mitigate overfitting.

The plot presented in Fig. 24 illustrates the model's accuracy on both training and validation data. Training accuracy rises steadily to almost 100 %, demonstrating the model's high capacity to identify patterns in the training set. The model learns the training data quickly, as evidenced by the high initial training accuracy (0.9968) in the first epoch. The validation accuracy (0.9846) is similarly high but slightly lower, reflecting the initial gap between training and unseen data. Over epochs 2 to 4, a slight decrease in training accuracy remains above 0.99, showing robust learning, with validation accuracy following a fluctuating trend but remaining high, indicating decent generalization. During epochs 5 and 6, a dip in training accuracy to 0.9925 corresponds with the increase in loss, highlighting potential overfitting. The validation accuracy drops slightly but stays above 0.975, indicating the model still performs well. In the final epochs 7 to 9, the training accuracy peaks at 0.9996, suggesting near-perfect learning on training data, while the validation accuracy remains high (0.9842) with minor fluctuations, showing stable performance on unseen data. The high training and validation accuracy confirms the model's capability in accurately classifying arrhythmias, a crucial requirement for clinical relevance. The stable, high validation accuracy demonstrates the model's potential for reliable performance in real-world applications, where it will encounter new ECG signals. The close connection between training and validation accuracy suggests that the model is not significantly overfitting, which makes it reliable for usage in real-world scenarios.

Fig. 25 presents an in-depth analysis of the classification performance for each class in the confusion matrix based on the results obtained from the training set. In Fig. 25, the numeric labels correspond to specific arrhythmia types: 0 represents 'N', 1 represents 'V', 2 represents 'A', 3 represents 'R', and 4 represents 'L'. The diagonal elements represent the right classifications, and incorrect classifications are represented by the off-diagonal elements. For most arrhythmia types, high values along the diagonal denote good classification performance. There appears to be some misunderstanding between some classes, which might be improved with more research or data augmentation. The model's accuracy in classifying most arrhythmia types is demonstrated by the confusion matrix, which is essential for a trustworthy diagnosis. Misclassifications may indicate similarities in the ECG patterns of certain arrhythmias, suggesting areas for further model refinement or

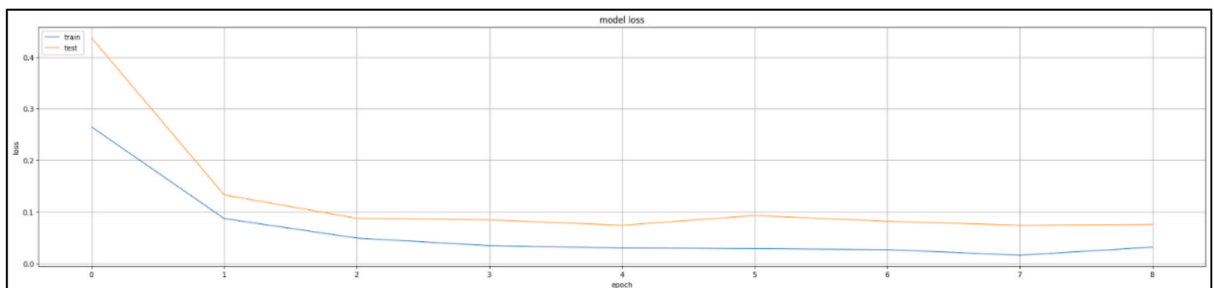


Fig. 21. Train vs. Test Loss for MS-CLDNet Model.

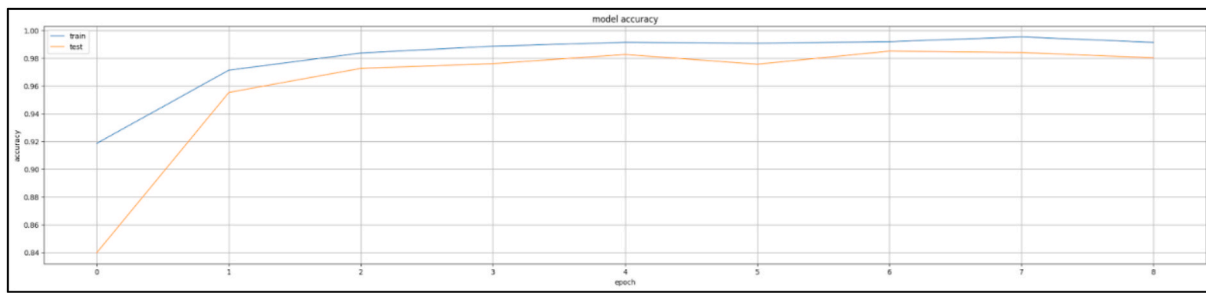


Fig. 22. Train vs. Test Accuracy for MS-CLDNet Model.

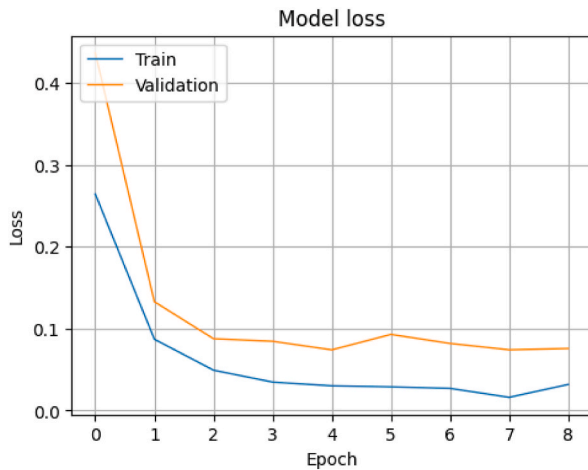


Fig. 23. Train vs. Validation Loss for MS-CLDNet Model.

additional feature extraction techniques.

AUC values close to 1.0 signify strong classification performance for every kind of arrhythmia, and the proximity of each curve to the upper-left corner signifies elevated specificity and sensitivity for the associated class. Fig. 27 displays the ROC curves for every class and shows the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity). AUC values that are high across classes attest to the model's resilience in differentiating between various arrhythmia kinds. The ROC curves reinforce the model's reliability for clinical applications, where both sensitivity and specificity are paramount for accurate arrhythmia detection.

The precision-recall curves presented in Fig. 28 shed light on the trade-offs that each class faces between recall and precision. High recall

and precision values show that the model is good at correctly identifying arrhythmias while reducing false positives, and the curves imply that the model strikes a good balance between recall and precision across various classes. The precision-recall analysis ensures that false positives and false negatives are kept to a minimum by confirming that the model functions well in situations where the prevalence of specific arrhythmias may be low. This balancing emphasizes how the concept can be applied practically in clinical situations where a mistaken diagnosis might have major consequences.

The thorough examination of the model's performance using a variety of plots shows that the MS-CLDNet model performs exceptionally well when it comes to the identification and categorization of arrhythmias on the MIT-BIH dataset. It is a great contender for use in clinical settings due to its high accuracy, dependable generalization ability, and strong classification performance. Future work can focus on further refining the model to address the observed fluctuations and misclassifications, ensuring even greater reliability and accuracy in real-world applications.

To present a more comprehensive evaluation of the model's performance, a detailed breakdown of key performance indicators for each class, including precision, recall, and F1-score, is given in Table 5, and a classification report is presented in Fig. 26. This complements the overall accuracy and provides a clearer picture of the model's robustness across all classes (see Table 6).

These results show that the model achieves a high degree of accuracy while also demonstrating strong performance in precision, recall, and F1-score across the different classes, with particular emphasis on its ability to classify the minority classes effectively.

Fig. 29 shows the training and validation accuracy plots that show how well our MS-CLDNet model performs at various K-Fold Cross-Validation folds. The reliability of high accuracy across folds for both training and validation sets highlights the model's ability to learn and generalize. This consistency gives users confidence in the model's classification of arrhythmia instances, which is a crucial component for

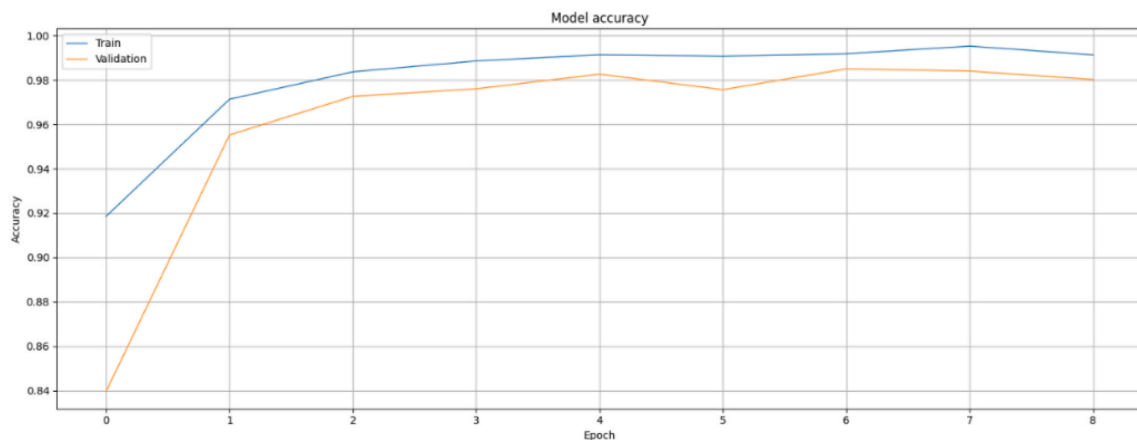


Fig. 24. Train vs. Validation Accuracy for MS-CLDNet Model.

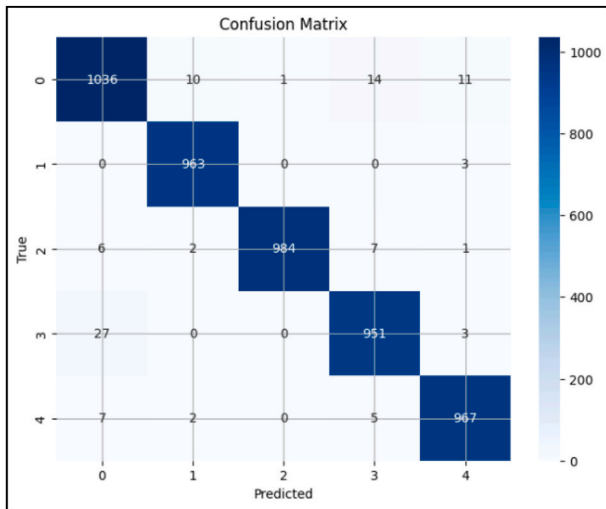


Fig. 25. Confusion matrix for MS-CLDNet model.

Table 5

Class Wise Evaluation Metrics for the proposed model.

Class	Precision	Recall	F1-Score	Support
Class 0	0.9830	0.9733	0.9781	1011
Class 1	0.9959	0.9949	0.9954	985
Class 2	0.9990	0.9746	0.9867	1025
Class 3	0.9520	0.9886	0.9699	963
Class 4	0.9941	0.9931	0.9936	1016

Table 6

Comparative analysis of arrhythmia detection methodologies.

S. No	Methodology	Accuracy	Overall Effectiveness and Concerns
1.	DenseNet Architecture [1]	97.56 %	Generalization concerns.
2.	Linear Discriminant Analysis (LDA) based classifier [4]	92.57 %	- Inconsistencies in classifier performance across different patients. - Low accuracy.
3.	2D CNN network architecture inspired by AlexNet [6]	94.7 %	Data limitation, Hardware constraints.
4.	A 12-layer CNN Arch was proposed. 60 epochs and batch size of 36. 10-fold cross-validation [7]	97.41 %	Limited dataset size and potential overfitting.
5.	CNN-LSTM [8]	97.98 %	Low classification accuracy for the atrial flutter (AFL) category, large computing costs.
6.	CNN-BiLSTM and DeepArr [combines feed-forward and recurrent neural networks] [10]	96.46 %	Limited training and testing database.
7.	Proposed MS-CLDNet	98.22 %	- High Accuracy. - Equipped to handle temporal patterns and refine features. - Noise Robustness (By employing wavelet-based denoising) - Low Loss values. - Integration of advanced neural architectures and efficient pre-processing techniques

	precision	recall	f1-score	support
0	0.98	0.97	0.98	1011
1	1.00	0.99	1.00	985
2	1.00	0.97	0.99	1025
3	0.95	0.99	0.97	963
4	0.99	0.99	0.99	1016
accuracy			0.98	5000
macro avg	0.98	0.98	0.98	5000
weighted avg	0.99	0.98	0.98	5000

Fig. 26. Detailed classification report for MS-CLDNet model.

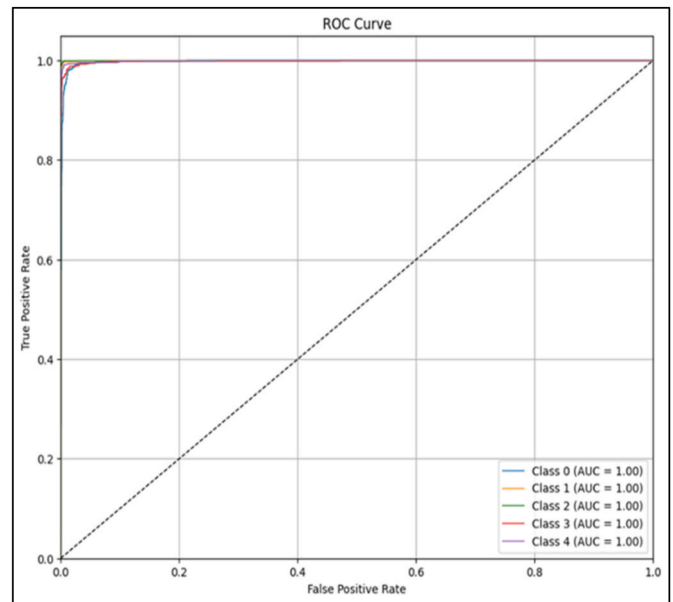


Fig. 27. Roc curve for MS-CLDNet model.

trustworthy arrhythmia detection in clinical settings. Analyzing the training and validation loss plots shown in Fig. 30 provides important information about our model's learning dynamics. Effective learning is indicated by a trend toward decreasing training and validation loss, and robust generalization is suggested by the minimal divergence between the two. These findings are essential for spotting possible overfitting and directing optimization tactics to guarantee the model's dependability and capacity for generalization in practical contexts. The training and validation accuracy and loss plots visually represent the model's learning process over the epochs. The model performs better on the training data with each epoch, as seen by the training accuracy plot, which increases steadily. In this paper, the test and validation results indicated that the model performs well across the different arrhythmia classes. However, some signs of overfitting were observed in the training loss curves. It was particularly evident as the model showed a slight improvement in performance on the training set but had relatively stable or slightly reduced performance on the validation set as training progressed. To address this, several strategies could be employed, such as regularization techniques (e.g., L2 regularization), the use of dropout layers to prevent the model from over-relying on specific features, and early stopping to halt training when performance on the validation set ceases to improve. Additionally, augmenting the training data or employing cross-validation during model evaluation could also help reduce the risk of overfitting, ensuring better generalization to unseen data.

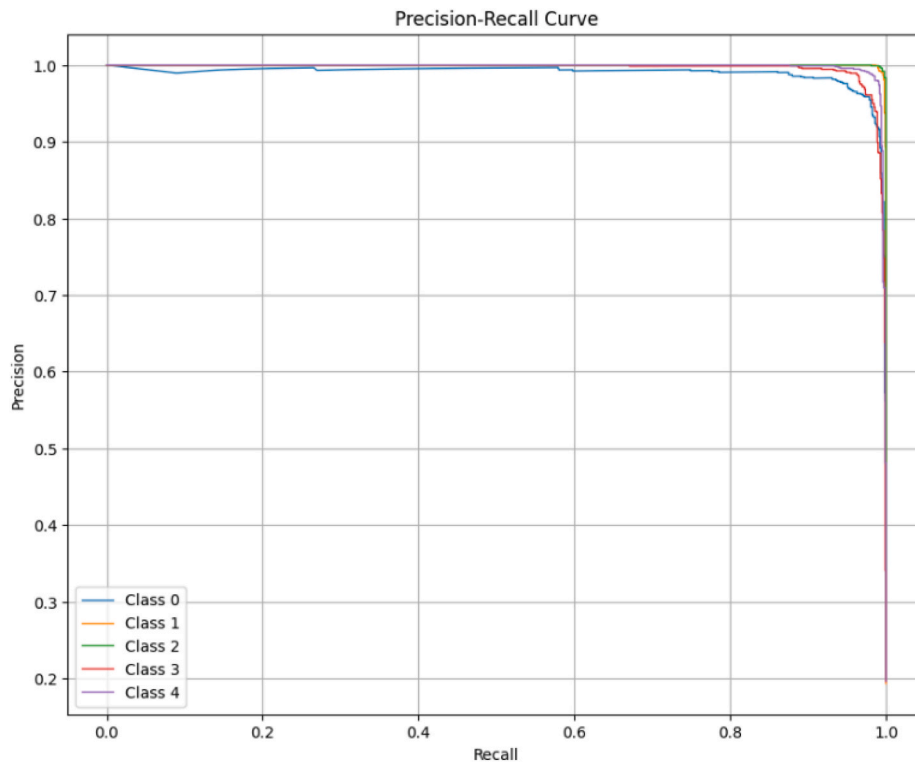


Fig. 28. Precision-recall curve for MS-CLDNet model.

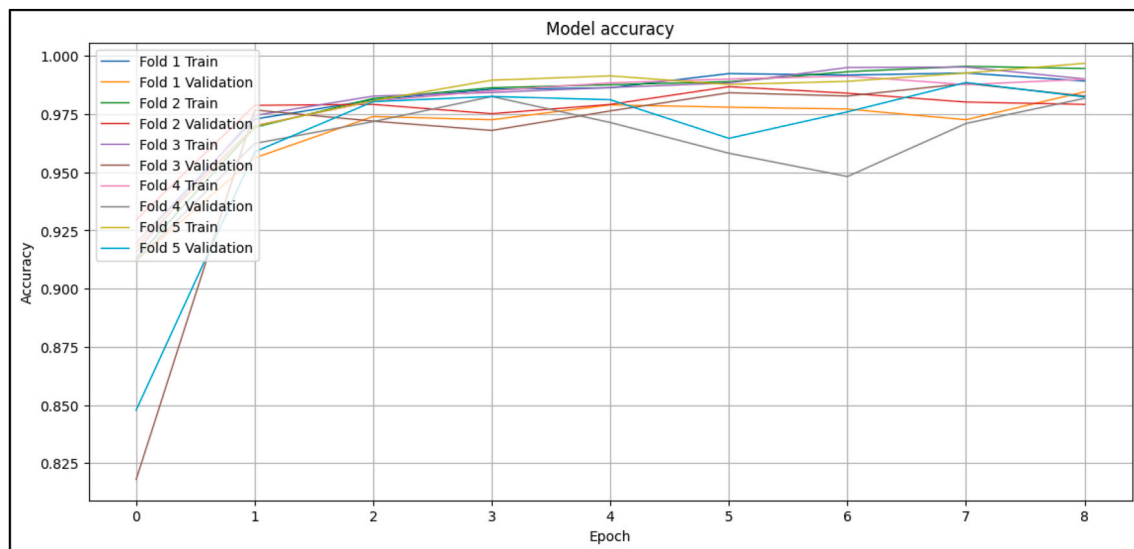


Fig. 29. Training and validation accuracy for each fold.

This pattern is also reflected in the validation accuracy plot, albeit with a few minor variations, indicating that the model generalizes well to new data. The accuracy curves for training and validation are nearly aligned, which highlights the model's capacity to sustain high performance without overfitting. The training and validation losses in the loss plots decrease with time, providing additional evidence of the model's learning effectiveness. The validation loss fluctuates from time to time. Still, these variations are expected and do not deviate significantly from the training loss, thereby validating the stability and dependability of the model.

The confusion matrix from the K-Fold Cross-Validation illustrated in Fig. 31 reinforces the model's strong discriminative capability on the

testing set. In Fig. 31, the numeric labels correspond to specific arrhythmia types: 0 represents 'N', 1 represents 'V', 2 represents 'A', 3 represents 'R', and 4 represents 'L'. High values along the diagonal (correctly classified instances) for each class (e.g., 958 for class 1) demonstrate the model's effectiveness in accurately identifying arrhythmia types. Conversely, minimal off-diagonal elements signify a low rate of misclassification between classes. This suggests the model generalizes well to unseen data within this fold, achieving a high degree of accuracy in differentiating various arrhythmias.

The ROC curves and corresponding AUC values from the K-Fold Cross-Validation process that is illustrated in Fig. 32 offer valuable insights. The ROC curves for each arrhythmia class (Class 0–4) reside close

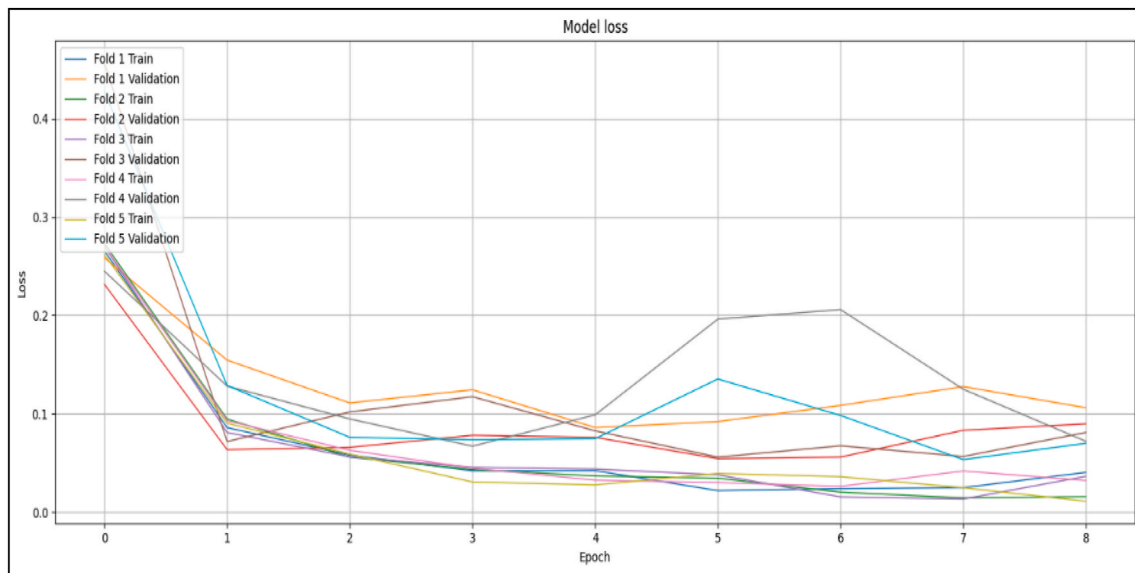


Fig. 30. Training and validation loss for each fold.

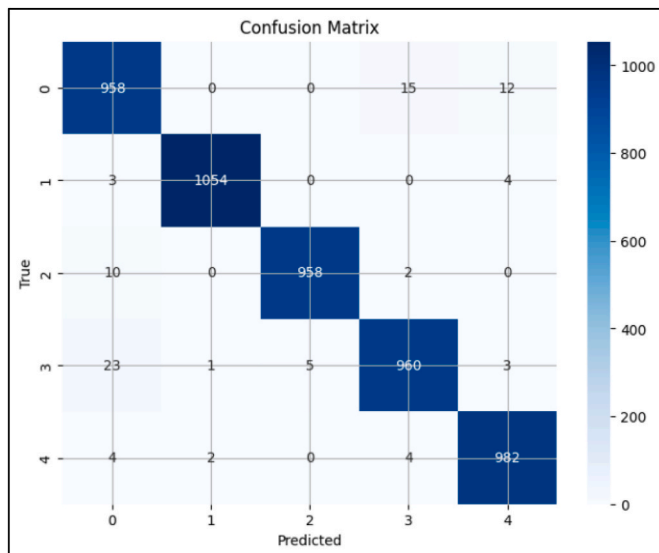


Fig. 31. Confusion matrix for the model after K-fold validation.

to the top-left corner of the graph illustrated in Fig. x, signifying exceptional model performance. This proximity to the ideal ROC curve shape indicates the model achieves both high sensitivity (correctly classifying true positive cases) and high specificity (correctly classifying true negative cases) across all classes. Furthermore, the AUC values substantiate this exceptional performance since all classes achieve AUC values very close to 1.0 (1.00 for Class 0, 1, 2, and 3; 0.99 for Class 4), signifying the model's remarkable ability to distinguish between different arrhythmia classes with exceptional confidence. In conjunction with the prior analysis of the confusion matrix, these ROC curves and AUC values provide compelling evidence that the MS-CLDNet model achieves robust and highly accurate classification of arrhythmias within the MIT-BIH dataset.

The model's performance is further validated by the precision-recall curves shown in Fig. 33, especially when it comes to imbalanced data classification. Precision shows the percentage of true positive predictions among all positive predictions (avoid false positives) on the y-axis. On the x-axis, recall shows how well the model captures all positive cases while avoiding false negatives. Our goal should be to achieve both

high recall and high precision. Each class (Class 0, Class 1, Class 2, Class 3, and Class 4) in the precision-recall curves shown in Fig. 33 has curves that show a good balance between precision and recall over a sizable portion of the curve's range. It suggests that the model can reduce the number of false negatives while simultaneously effectively differentiating between true positive cases and false positives. This feature is especially important when working with unbalanced datasets, as some arrhythmia classes may be more common than others. The model shows its dependability in correctly identifying true positive arrhythmia instances while maintaining low false positive and false negative rates by attaining high precision and recall across all classes. This discovery strengthens the model's excellent performance in classifying arrhythmias and adds to the encouraging findings from the ROC curve and confusion matrix analyses.

Analyzing the loss versus accuracy plots for the training and validation sets across all folds of the K-Fold Cross-Validation process was done to obtain a thorough understanding of the model's learning behavior during the training process. These plots, as shown in Fig. 34, provide important information about how well the model learns from the training set and generalizes to new data. Four curves are usually included in the plots: training accuracy, validation accuracy, training loss, and validation accuracy. As the model gains proficiency in minimizing the loss function on the training data, the training loss should decline gradually. As the model performs better on training data, the training accuracy has to rise together. As time goes on, the validation loss should also drop, suggesting that the model can be applied to data that has not been seen before. However, because different validation sets are used in each fold, the validation loss curve may show fluctuations when compared to the training loss curve. For the same reason, the validation accuracy may exhibit slight variations even though it should ideally rise together with the training accuracy. The following patterns were seen in the MS-CLDNet model's loss versus accuracy plots.

- **Training Loss and Accuracy:** Effective learning was indicated by a training loss that started higher and gradually dropped over epochs. On the other hand, the training accuracy started out lower and grew over time, indicating the model's increasing capacity to identify arrhythmias in the training set accurately.
- **Validation Loss and Accuracy:** The validation loss showed a declining trend over epochs with some fluctuations, matching the training loss's pattern. With only slight fluctuations over time, the

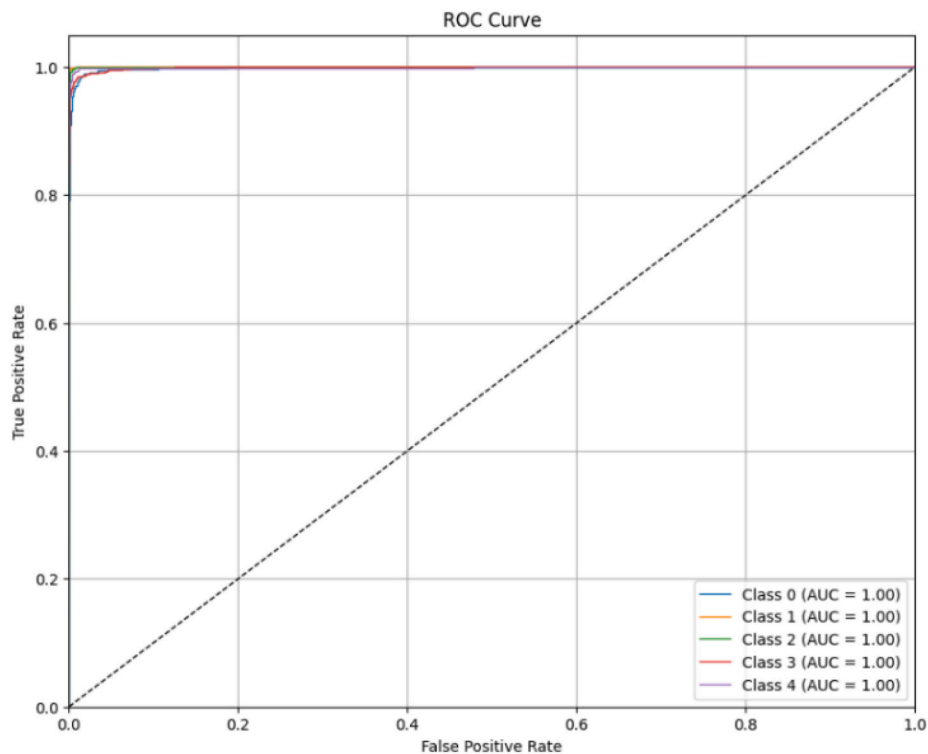


Fig. 32. ROC curve for the model after K-fold validation.

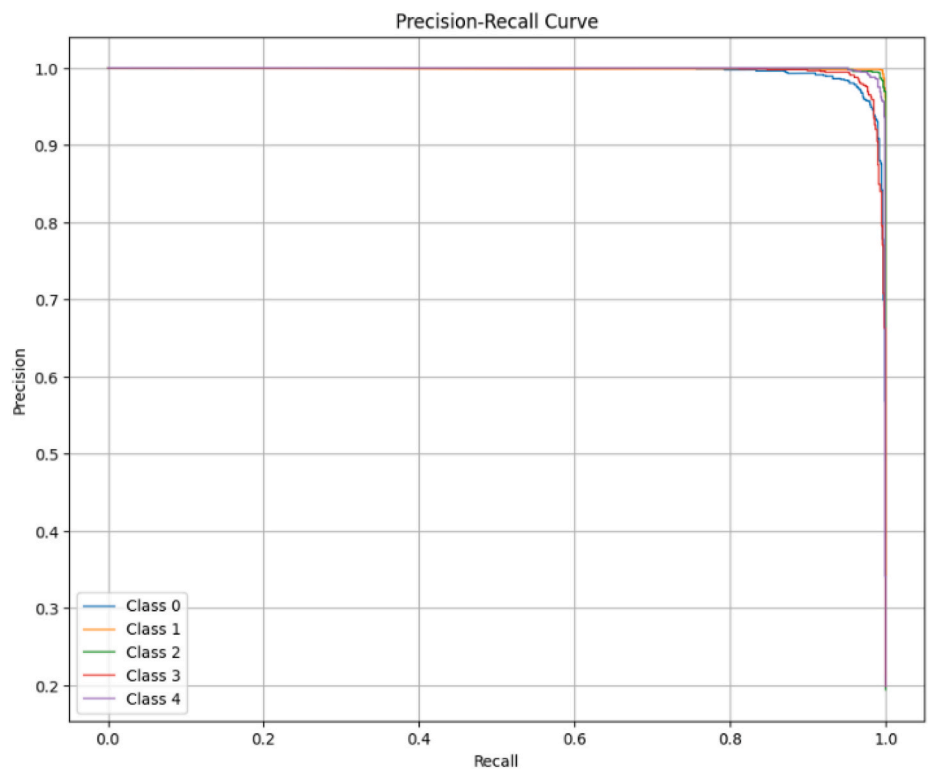


Fig. 33. Precision-recall curve for the model after K-fold validation.

validation accuracy also rose. These findings imply that the model is effectively generalizing to previously unseen data within each fold.

The convergence of the training and validation loss curves indicates how well the model minimizes the loss function. The minimal gap

between the training and validation accuracy curves further supports the model's capacity to generalize well on unfamiliar data and prevent overfitting. The consistency of these data across all folds supports the model's resilience in classifying arrhythmias.

The distribution of expected probability among various arrhythmia

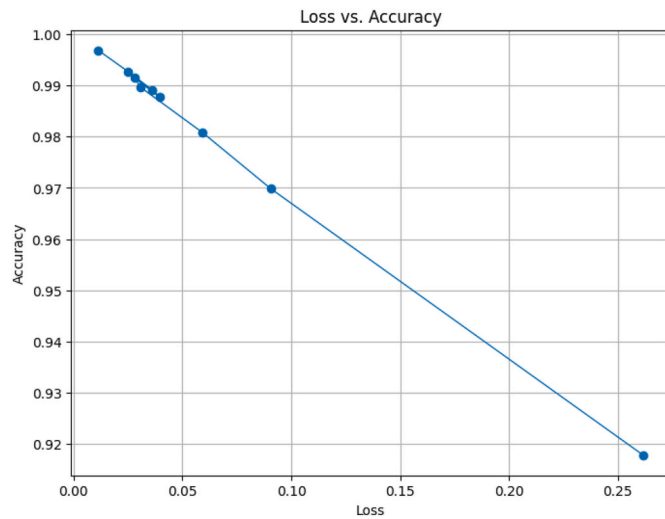


Fig. 34. Loss vs accuracy plot for the model after K-fold validation.

classes can be understood by examining the class probabilities vs. frequency plots shown in Fig. 35. These graphs show the model's confidence in its predictions for each class and were produced following the application of 5-fold cross-validation. A histogram of the expected probabilities for a particular arrhythmia class is shown in each subplot. The distribution of these probabilities offers key information about the model's prediction behavior. In the provided plots, we observe a concentration of frequency at probability values near 0 and 1. It indicates that the model frequently assigns high confidence scores (close to 0 or 1) to its predictions, signifying a strong belief in the correctness of its classifications. Conversely, there are fewer predictions with intermediate probability values, suggesting that the model rarely outputs ambiguous classifications that fall around the 0.5 mark. This distribution of predicted probabilities reinforces the model's proficiency in differentiating between various arrhythmia classes. It looks like the multi-scale CNN-LSTM-Dense architecture is successful in identifying the underlying patterns in the data, enabling the model to produce resolute and confident predictions. These observations align with the high accuracy results achieved during the K-fold validation process, further corroborating the model's overall effectiveness in arrhythmia

classification.

During the K-fold cross-validation procedure, the training and validation performance metrics are shown in Figs. 36 and 37—were gathered over five folds. The model was trained for nine epochs for each fold, and the corresponding training and validation loss and accuracy were noted. The learning process was regularly assessed using the training loss and accuracy metrics, and the model's generalization performance on unobserved data was evaluated using the validation metrics.

All epochs of the training loss trended downward, suggesting that the model was successfully learning from the training set. For example, in fold 1, training accuracy increased from 91.97 % to 98.93 %, but training loss decreased from 0.2656 in the first epoch to 0.0406 in the ninth epoch. This pattern was also seen in the other folds, indicating that the model can gradually lower errors and enhance its predictions on the training set. The validation loss generally demonstrated a decreasing

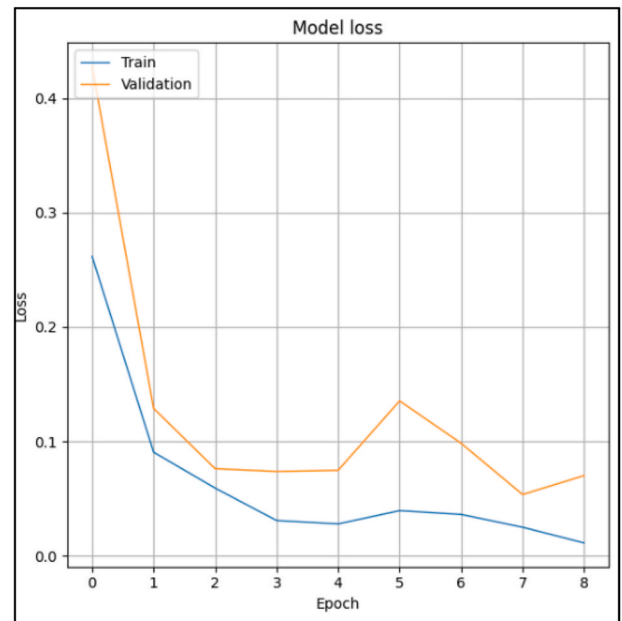


Fig. 36. Overall training and validation loss values for the model after K-fold validation.

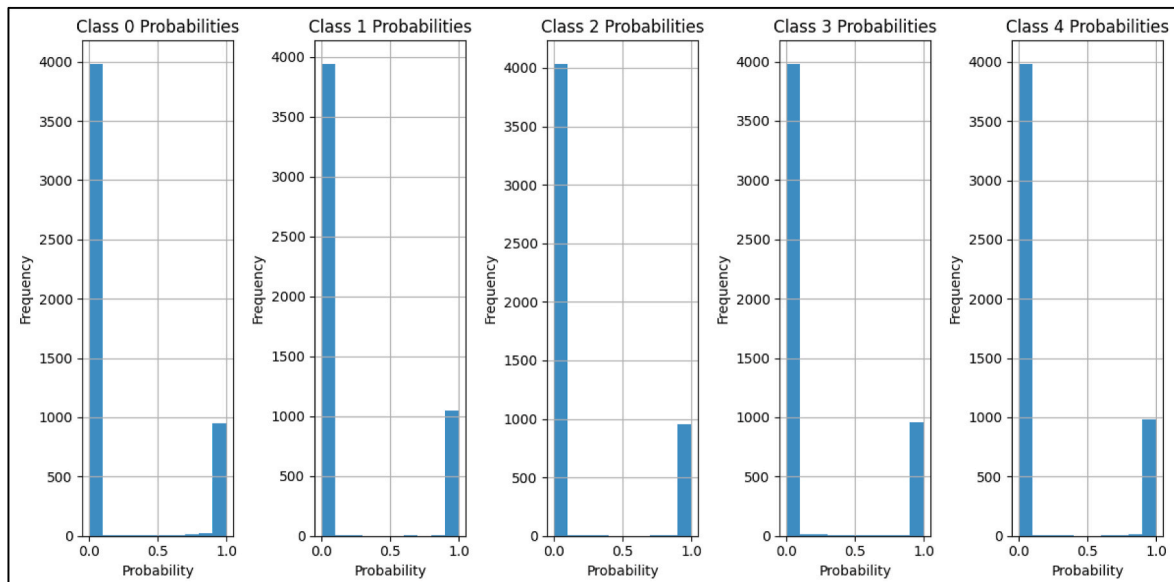


Fig. 35. Class Probabilities vs Frequency plots for the model after K-fold Validation.

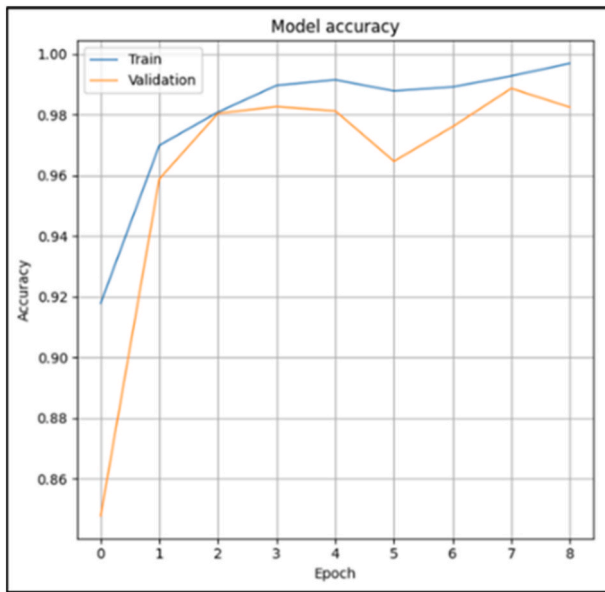


Fig. 37. Overall training and validation loss values for the model after K-fold validation.

trend across epochs despite fluctuating more than the training loss, indicating that the model was not only learning but also generalizing well. For instance, in fold 1, the validation loss started at 0.2585 in the first epoch and decreased to 0.1062 by the ninth epoch, while the validation accuracy increased from 91.22 % to 98.46 %. Similar patterns were observed in the other folds, confirming the model's robustness and consistency. The average accuracy across all five folds was 98.22 %, with a standard deviation of 0.17 %, indicating high performance and low variance, thus demonstrating the reliability and stability of the model across different subsets of the data. The average loss was 0.0839, reflecting the model's efficiency in minimizing errors during predictions.

Analyzing the learning rate dynamics of our MS-CLDNet model involves examining the scheduled changes over epochs and their influence on crucial training metrics like accuracy and loss. The learning rate schedule implemented utilizes a custom 'LearningRateScheduler' function with an initial steady rate for the first three epochs, followed by an exponential decay strategy for subsequent epochs. This strategy is designed to balance exploration and fine-tuning of model parameters, aiming to prevent premature convergence and mitigate overfitting. Throughout the K-Fold Cross-Validation process ($K = 5$ folds), each fold independently trains the model while tracking learning rate values via the 'LrHistory' callback. This approach facilitates a clear visualization of how the learning rate evolves over epochs, typically showing a progressive decline post the initial epochs, aligning with the planned exponential decay. The plot depicted in Fig. 38 shows that the initial phase employs a steady learning rate for the first three epochs, allowing for exploration of the parameter space.

Subsequently, an exponential decay strategy takes effect, gradually reducing the learning rate in the following epochs. The goal of this approach is to achieve a balance between parameter tuning and exploration. The initial steady rate allows the model to explore different regions of the parameter space and potentially identify promising areas for convergence. The following exponential decay helps the model refine its parameters by focusing on areas with better performance, ultimately mitigating the risk of premature convergence and overfitting. The learning rate schedule promotes both effective learning and generalization by facilitating exploration in the initial stages and then guiding the model towards optimal parameter values through decay, ultimately contributing to the model's robust performance observed in the k-fold

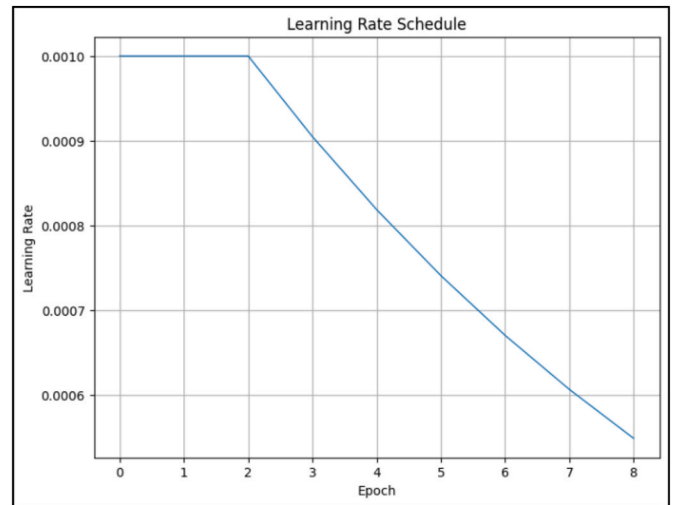


Fig. 38. Learning Rate Schedule for the model.

cross-validation process.

The K-fold Cross-Validation process of the MS-CLDNet model yields the model accuracy plot shown in Fig. 39, which is an essential visual aid for evaluating the model's generalization and learning dynamics. This plot provides detailed insights into the model's performance over the training iterations by capturing two key curves over epochs: training accuracy and validation accuracy. The training accuracy curve shows a steady upward trajectory during the training process, meaning that the model gradually improves its parameters and picks up on the complex patterns found in the training data. The model's capacity to accurately represent and adjust to the intricacies present in arrhythmia classification tasks is demonstrated by this consistent improvement. Meanwhile, with only slight variations, the training accuracy curve's trends are reflected mainly in the validation accuracy curve. This coherence shows that, for each cross-validation fold, the model generalizes well to fresh, untested data. The model exhibits remarkable performance on the training set, but it also sustains strong performance on the validation set. It suggests that the model can generalize beyond the particular examples it was trained on, as evidenced by the small differences in the validation and training accuracies. One important result from the accuracy plot is the closeness of the training and validation accuracy curves over different epochs. This alignment demonstrates the effectiveness of the learning rate schedule included in the model. The learning rate schedule makes sure that the model learns as best it can without overfitting (memorizing the training data) or underfitting (failing to capture important patterns) by skilfully balancing the exploration and exploitation phases of learning. This coherence in accuracy curves substantiates the model's capacity to achieve high-performance metrics consistently across diverse folds and datasets. The consistency in training accuracy improvement and validation accuracy maintenance, as depicted in the model accuracy plot, corroborates earlier findings from the K-Fold Cross-Validation analysis. This convergence of results underscores the model's reliability and robustness in accurately classifying arrhythmia instances, which is crucial for its application in clinical settings where precision and reliability are paramount. The model accuracy plot serves as a compelling visual testament to the MS-CLDNet model's efficacy in learning from data and generalizing to new instances. Its potential as a dependable tool for real-world arrhythmia detection and diagnosis is validated by its ability to maintain high-performance metrics across multiple folds, confirming its relevance and applicability in clinical practice. Adjustments based on these insights could further optimize the model's performance, enhancing its suitability for broader clinical datasets and applications.

The most obvious finding from the loss curves shown in Fig. 40 is that

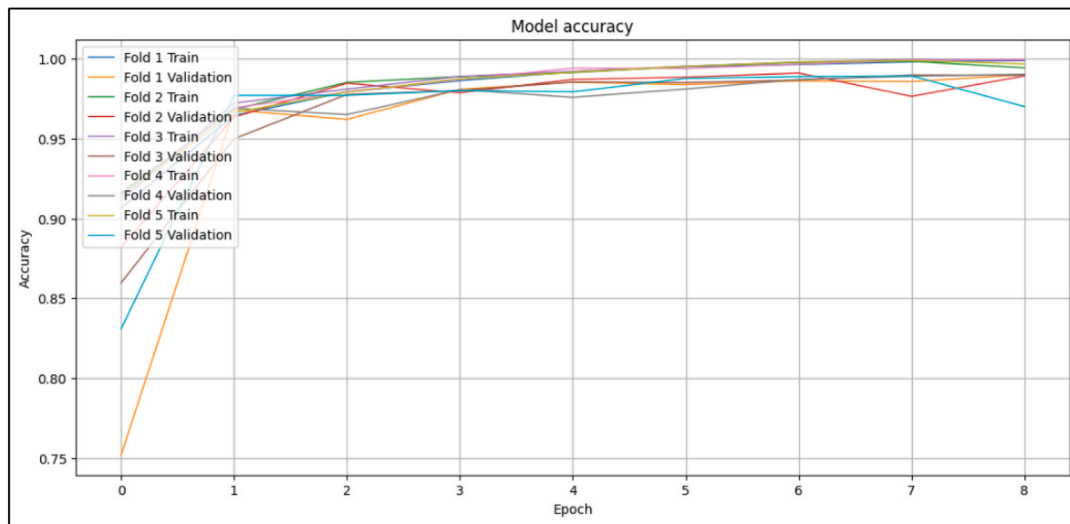


Fig. 39. Model Accuracy for each epoch after K-fold Cross-Validation.

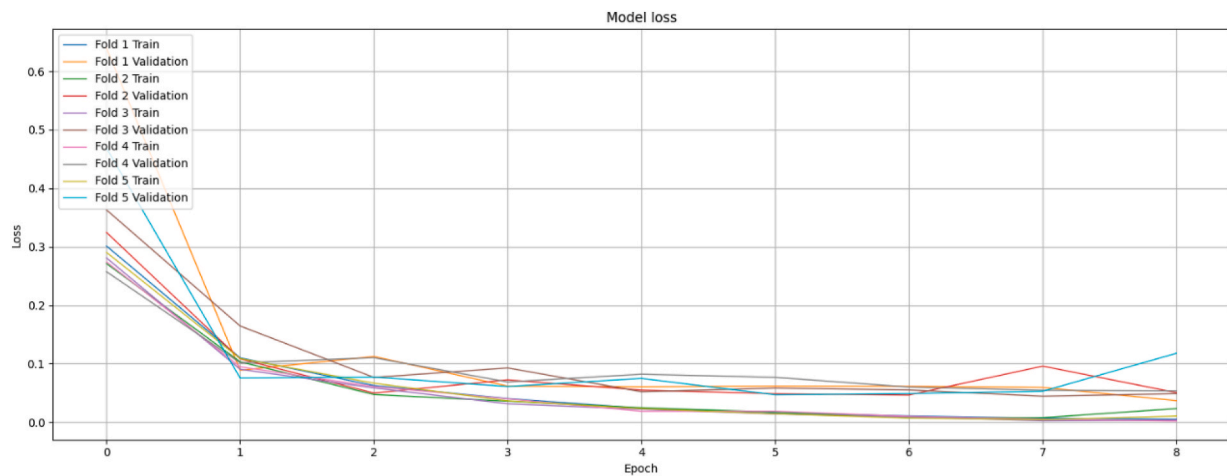


Fig. 40. Model Loss for each epoch after K-fold Cross-Validation.

they always have a downward trajectory over the course of the epochs. As the model iteratively modifies its parameters, the training and validation loss curves both show a progressive decline, suggesting that the model continuously increases its predictive accuracy. This decrease in loss indicates that the model learns to predict future events accurately and successfully extracts the underlying patterns from the training set. Additionally, the validation loss curve is relatively stable and resembles the training loss curve. This alignment shows that the model performs well in terms of generalizing to fresh, untested data—an important aspect validated by the K-Fold Cross-Validation process. Although modest differences in validation loss may occur due to various validation sets in each fold, the general stability between training and validation loss illustrates the model's resiliency in maintaining constant performance metrics over a variety of datasets. The coherence between the training and validation loss curves further demonstrates the efficacy of the learning rate schedule incorporated into the model. Optimizing model training is made easier with a well-balanced learning rate schedule, which first explores larger parameter spaces at higher rates before progressively narrowing parameters to avoid overfitting. The model can learn effectively without overfitting the training set or underperforming on fresh data if the learning rate schedule maintains this balance, as indicated by the small gap between these curves. The K-fold Cross-Validation analysis results and the patterns in the loss curves

agree fairly well. The model's consistent ability to learn and effectively decrease loss on a range of datasets is validated by the trend toward reducing training and validation loss across folds. The stable validation loss further corroborates the model's capacity to generalize well, demonstrating reliable performance across different subsets of data. Overall, the loss curves presented provide compelling evidence of the MS-CLDNet model's robust learning dynamics and stable performance metrics. The model's ability to learn from data and retain generalization to new instances is demonstrated by the constant loss decrease that occurs during training and the low divergence between the training and validation loss curves.

The confusion matrix, ROC curves, and AUC scores provided in Figs. 41 and 42 give comprehensive insights into the discriminative performance and classification accuracy of our MS-CLDNet model in arrhythmia detection. These metrics were evaluated across folds of K-Fold Cross-Validation, complementing the analysis of learning rate dynamics. ROC curves visually depict the trade-offs between sensitivity and specificity. It is crucial to understand how well our model distinguishes between arrhythmia classes across varying thresholds set by the learning rate schedule. A higher AUC score indicates superior discriminative ability directly influenced by the model's learning rate adjustments during training. Concurrently, confusion matrices offer a detailed breakdown of classification errors, highlighting where the model's

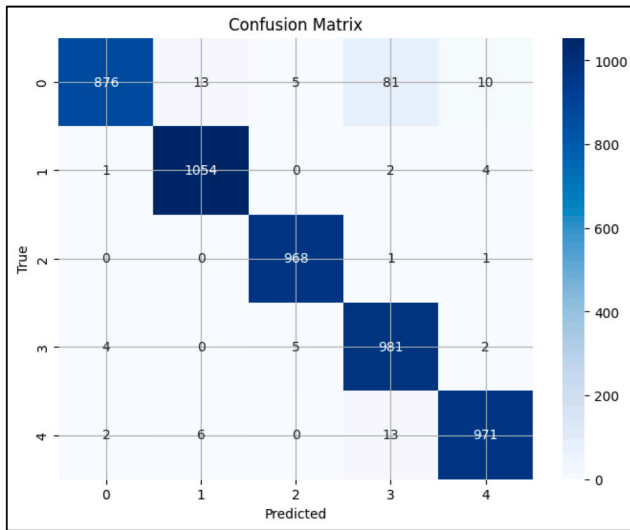


Fig. 41. Confusion matrix after K-fold cross-validation and analysis of learning rate dynamics.

predictions align or deviate from actual arrhythmia types. The alignment of these metrics with learning rate dynamics underscores their interdependence in optimizing model performance. By iteratively adjusting learning rates based on ROC curves, AUC scores, and confusion matrices, we fine-tune our model to achieve robust and accurate arrhythmia classification. This integrated approach not only validates the effectiveness of our learning rate strategy in promoting model generalization and discrimination but also enhances the model's utility in real-world clinical settings, where precise classification of arrhythmias is paramount.

A classification report provides critical insights into a model's predictive accuracy across various classes by thoroughly assessing the model's performance in multi-class classification. Precision highlights

the model's capacity to prevent false positives by comparing accurate positive predictions to all predicted positives. Recall, on the other hand, evaluates how well the model captures the most positive instances by accurately identifying each one. Models with balanced performance in both areas are favored by the F1 score, which integrates precision and recalls into a single metric. Support provides context on the class distribution by indicating the number of instances per class. The tabular format of the classification report facilitates rapid comparisons between classes and helps direct optimization efforts, such as hyperparameter tuning, to improve overall model reliability in real-world applications. According to the classification report shown in Fig. 43, the model performs admirably in each of the five arrhythmia classes (Class 0 through Class 4). The report shows high recall, F1-score, and precision values for every class, which demonstrate the model's overall balanced performance across all classes, accuracy in predicting positives, and capacity to identify true positives accurately. Class 1 stands out due to its robust classification capability, as evidenced by its exceptionally high recall (0.98), F1 score (0.99), and precision (0.99). Despite having slightly lower metrics than the other classes, classes 3 and 4 nevertheless have F1 scores above 0.94, demonstrating a good balance between recall and precision. Examining the "Support" values in the report reveals varying class sizes within the test data, with Class 0 having the highest support (1059 instances) and Class 4 the lowest (139 instances). This distribution suggests a potential class imbalance, where certain arrhythmia types are less represented than others. Despite this imbalance, the model consistently performs well across all classes, demonstrating its generalizability and effectiveness in handling diverse data distributions. The results underscore the model's strengths in accurate arrhythmia classification, which is particularly noteworthy given the dataset's varying frequencies of different arrhythmia types.

5. Conclusion and discussion

Using the MIT-BIH dataset, this paper thoroughly evaluates the Multi-Scale CNN-LSTM-Dense model for arrhythmia detection. Class rebalancing techniques have been incorporated to improve model

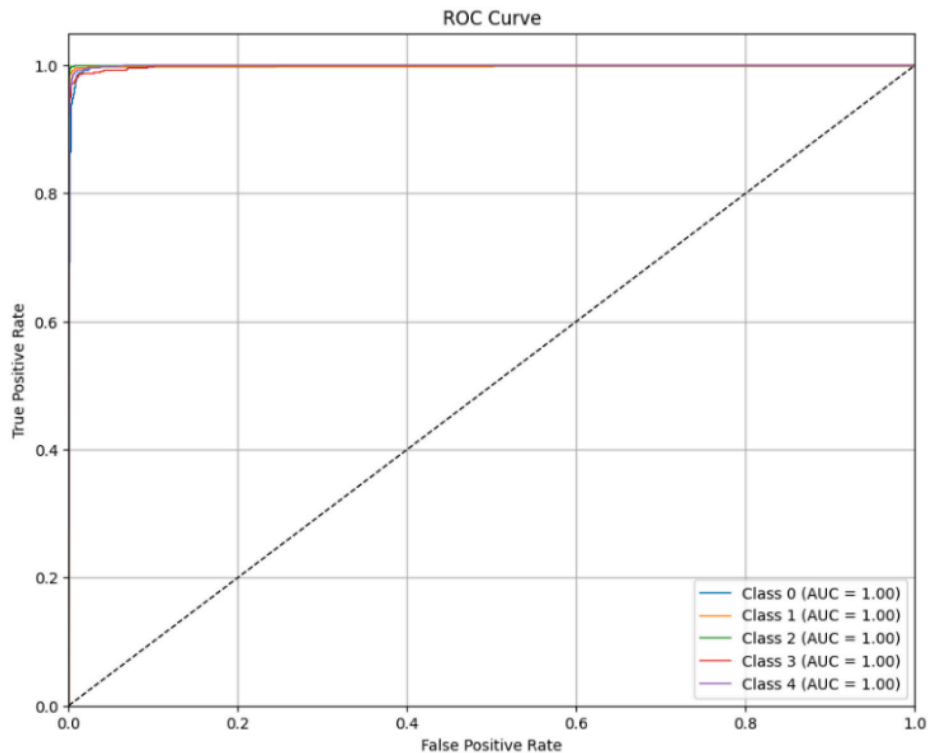


Fig. 42. Roc curve after K-fold cross-validation and analysis of learning rate dynamics.

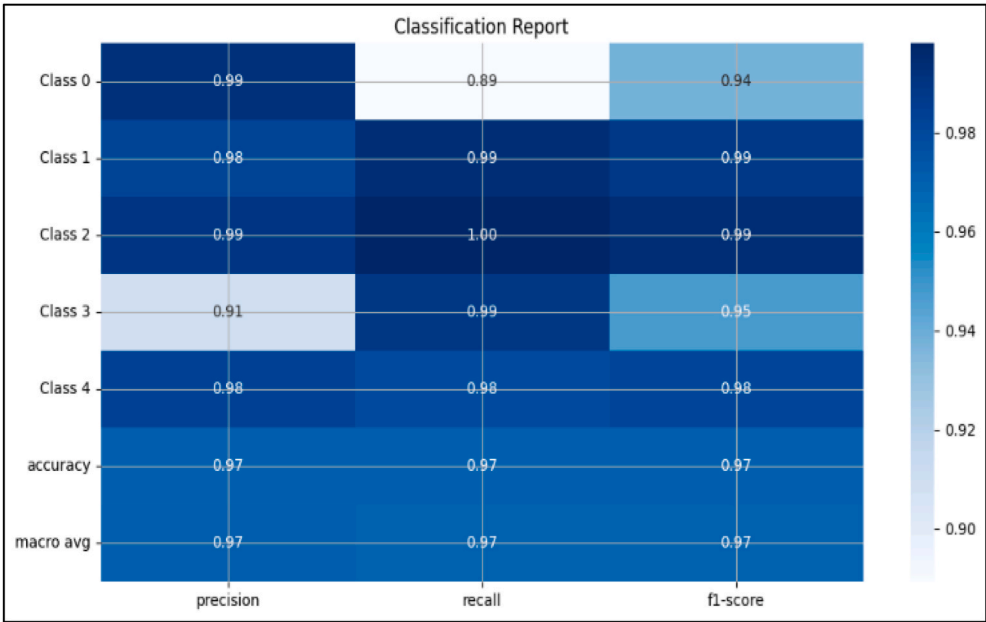


Fig. 43. Classification report for the model after K-fold cross-validation and analysis of learning rate dynamics.

performance. An average of 98.22 % with slight variance is consistently shown by rigorous five-fold cross-validation, indicating high accuracy. Low average loss values (0.084), which show exact alignment between expected and actual outcomes, validate the robustness of the model. These findings highlight the approach’s stability and dependability in correctly categorizing arrhythmias, which is crucial for trustworthy clinical decision-making. An organized method of training models is revealed by analyzing learning rate dynamics, which strikes a balance between parameter space exploration and exponential decay fine-tuning. With the help of this tactic, the model can efficiently and without overfitting identify intricate patterns in the data. The training and validation accuracy curves’ alignment over epochs shows how adaptable the model is to novel, untested data, which is crucial for practical applications in healthcare settings. Deeper insights into the discriminative power of the model can be gained from visual evaluations like confusion matrices, ROC curves, and precision-recall curves. Superior sensitivity and specificity in identifying arrhythmias are demonstrated by high AUC values in ROC curves that are nearly 1.0 in all classes. Precision-recall curves also demonstrate how the model can reduce false positives and false negatives, which is essential for precise clinical diagnosis.

In terms of clinical applicability, the model’s potential to assist in real-time arrhythmia classification can be realized through its integration into existing ECG screening workflows. By incorporating the model into ECG monitoring systems, it could function as a real-time classifier, alerting healthcare professionals to the presence of arrhythmias, thus enabling quicker diagnosis and intervention. The model could be deployed on portable ECG devices or wearable ECG patches, facilitating continuous monitoring for patients at risk of arrhythmias, such as those with cardiovascular diseases. Furthermore, integrating the model into clinical decision support systems (CDSS) would allow clinicians to receive actionable insights and recommendations based on real-time ECG data, helping in the early detection of arrhythmias and the formulation of appropriate treatment plans. Moreover, the model’s integration into telemedicine platforms would be beneficial for remote monitoring, especially in underserved areas where access to healthcare may be limited. These proposed integrations would not only enhance diagnostic accuracy but also ensure timely intervention, improving patient outcomes by facilitating continuous monitoring, early detection, and personalized care.

While the model integrates complex architectures such as DenseNet and Long Short-Term Memory (LSTM) networks, these modules introduce significant computational costs, which may challenge their deployment on resource-constrained devices. DenseNet’s densely connected convolutional layers can be computationally expensive due to high memory and processing demands. At the same time, LSTMs add to the computational overhead due to their sequential nature and long-term memory processing requirements. In resource-constrained environments, such as real-time systems with limited CPU/GPU power, memory, and storage, these factors could hinder practical deployment by slowing inference times and increasing resource consumption. Strategies such as model pruning, quantization, knowledge distillation, and leveraging edge computing with hardware acceleration could help optimize the model for deployment on such devices to address these challenges. Future research will explore these techniques to improve the model’s efficiency and its suitability for real-time, resource-limited systems.

This research highlights the significant advancements achieved with the proposed Multi-Scale Convolutional LSTM Dense Network (MS-CLDNet) for cardiac arrhythmia classification, which attained a remarkable accuracy of 98.22 % on the MIT-BIH arrhythmia dataset, accompanied by a low average loss of 0.084. This performance surpasses existing methods in the literature, such as traditional machine learning algorithms like Support Vector Machines, which typically yield accuracies around 90 %, and other recent studies using Convolutional Neural Networks that report around 94 % accuracy but fail to address class imbalance adequately. The integration of bidirectional Long Short-Term Memory (LSTM) networks enables the model to capture complex temporal patterns in ECG signals, enhancing its ability to detect subtle arrhythmia variations. At the same time, the use of Dense Blocks refines features for improved classification accuracy. Despite these strengths, the study acknowledges limitations, particularly regarding performance on datasets with varying noise levels. It suggests future work could explore real-time ECG monitoring applications and expand the dataset for greater diversity in arrhythmia types. Overall, the MS-CLDNet model represents a significant step forward in automated arrhythmia detection, promising to enhance early diagnosis and improve patient outcomes in cardiac healthcare.

6. Future work

Future research will focus on implementing the MS-CLDNet model in real-time ECG monitoring systems. Evaluating its performance in a clinical setting will provide insights into its practicality and effectiveness for continuous arrhythmia detection. Further exploration into advanced noise reduction techniques will be beneficial. Investigating the model's performance on ECG signals with varying levels of noise and interference will enhance its applicability in real-world scenarios where signal quality can be compromised. It could also involve integrating ECG data with other physiological signals (e.g., heart rate variability, blood pressure) to develop a more comprehensive cardiovascular monitoring system. This multimodal approach may improve the accuracy and reliability of arrhythmia detection. Researching methods to optimize the model for deployment on resource-constrained devices, such as wearable technology or mobile applications, will facilitate broader access to advanced arrhythmia detection tools in everyday healthcare settings.

In future work, the aim is to expand the scope of our model by incorporating additional physiological data, such as heart rate variability (HRV) and blood pressure, alongside ECG signals. Integrating these complementary data sources could enhance the model's predictive accuracy and robustness, providing a more holistic approach to arrhythmia classification. Exploring the impact of multi-modal data will be a key step toward improving the clinical relevance and applicability of our model in real-world settings.

To provide actionable directions for real-time deployment and mobile integration, the focus is on several key steps. First, work on optimizing the model is crucial to reduce its computational complexity, employing techniques such as model pruning, quantization, and knowledge distillation. It will ensure the model can run efficiently on resource-constrained devices. Next, focus on optimizing inference speed to provide real-time performance, including strategies for reducing latency and improving processing time on mobile platforms. Additionally, it focuses on the development of a user-friendly mobile application with an intuitive interface, enabling healthcare professionals to interact with the model easily. The application will include features for real-time arrhythmia classification, with visualizations of results and clear decision support for clinicians. Lastly, the focus would be on conducting extensive testing to evaluate the model's real-time performance and ensure its reliability across diverse devices and clinical environments. These steps will facilitate the model's seamless integration into clinical practice, making it accessible for real-time use in arrhythmia detection.

CRediT authorship contribution statement

Kishor Kumar Reddy C.: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Advaita Daduvy:** Writing – original draft, Methodology, Data curation, Conceptualization. **Vijaya Sindhoori Kaza:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Mohammed Shuaib:** Writing – review & editing, Visualization, Resources, Methodology, Formal analysis, Data curation. **Muhammad Mohzary:** Writing – original draft, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Shadab Alam:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology. **Abdullah Sheneamer:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis.

Informed consent statement

Not applicable.

Dataset details

The MIT-BIH Arrhythmia Database has 48 half-hour segments of

two-channel ambulatory ECG recordings collected from 47 participants, make up the MIT-BIH Arrhythmia Database. Over a 10-mV range, the recordings were digitized at 360 samples per second per channel with 11-bit resolution. With annotations for arrhythmias and signal quality, this dataset offers a thorough resource for researching and analyzing cardiac arrhythmias. The dataset is a commonly used reference in the fields of arrhythmia detection and cardiovascular research.

Data availability

The data utilized in this study are publicly available in the MIT-BIH Arrhythmia Database, which can be accessed via PhysioNet (<https://physionet.org/content/mitdb/>). Researchers can download the dataset and associated documentation directly from the PhysioNet repository, ensuring transparency and reproducibility of the research findings. The MIT-BIH Arrhythmia Database used is also available at the following link: <https://www.kaggle.com/datasets/taejoongyoon/mitbit-arrhythmia-database>.

Ethical approval

This paper involves the use of publicly available anonymized data. The MIT-BIH Arrhythmia Database is freely accessible for research purposes and has been de-identified to protect patient privacy. All data handling and analysis procedures adhered to ethical standards and guidelines for the use of such datasets.

Institutional review board statement

Not applicable.

Ethical Statement

Hereby, I/**Muhammad Mohzary**/consciously assure that for the manuscript/A **Multi-Scale** Convolutional LSTM-Dense Network for Robust Cardiac Arrhythmia Classification from ECG Signals/the following is fulfilled:

- 1) This material is the authors' own original work, which has not been previously published elsewhere.
- 2) The paper is not currently being considered for publication elsewhere.
- 3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
- 4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
- 5) The results are appropriately placed in the context of prior and existing research.
- 6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.
- 7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.
- 8) Informed consent is not required as the article uses a publicly available OASIS Dataset Available online: <https://sites.wustl.edu/oasisbrains/home/oasis-3/>

The violation of the Ethical Statement rules may result in severe consequences.

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I agree with the above statements and declare that this submission follows the policies of Solid State Ionics as outlined in the Guide for Authors and in the Ethical Statement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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